Safety and immunogenicity of the recombinant adjuvanted herpes zoster vaccine Shingrix® in children and adolescents with acute lymphoblastic leukaemia (ALL):

A Phase 1/2 clinical trial

TSHINE

CLINICAL TRIAL ON MEDICINAL PRODUCT FOR HUMAN USE

Version N°1.0 dated ~~17/11/2025~~16/07/2026

Project Code: APHP251235 / EU CT number: 2025-524818-28-00

Commenté [SE1]: @Christine : A compléter avec nouveau numéro ?

Commenté [PC2R1]: En cours de mon côté

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SIGNATURE page for a research PROTOCOL

Research code number: APHP251235

Title: Safety and immunogenicity of the recombinant adjuvanted herpes zoster vaccine Shingrix® in children and adolescents with acute lymphoblastic leukaemia (ALL): a phase 1/2 clinical trial

Version no.1.0 dated: 16/07/2026

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:

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ABBREVIATIONS / GLOSSARY

Commenté [CPE4]: Il manque beaucoup d'abréviations ici qui sont dans le corps du protocole

AE: Adverse Event

AGEPS: *Agence Générale des Equipements et Produits de Santé*

AL: acute leukaemia

ALT: ALanine aminotransferase

AST: ASpartate aminotransferase

ALL: acute lymphoblastic leukaemia

CRM : Continual Reassessment Method

DLT: Dose Limiting Toxicity

DLT: Dose Limiting Toxicity

gE: glycoprotein E

GMT

HBV: Hepatitis B Virus

HCV: Hepatitis C Virus

HIV: Human Immunodeficiency Virus

HM: haematological malignancies

HSC : Hematopoietic Stem Cell

HZ/su: Herpes Zoster/subunit

HZ/su: Herpes Zoster/subunit

IFN: Interferon

IgG : Immunoglobulines G

IUD: intrauterine device

IUD: intrauterine device

IUS: intrauterine system

MGC

MGT

MTD: Maximum Tolerated Dose

MTD: Maximum Tolerated Dose

NK: Natural Killer

PNN : Polynucléaires neutrophiles

SAE: Serious Adverse Event

SAE: Serious Adverse Event

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SUSARs: suspected unexpected serious adverse reactions

TRV

VTR

VZV: varicella-zoster virus

VZV: varicella-zoster virus

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1. SUMMARY

Full title	Safety and immunogenicity of recombinant adjuvanted herpes zoster vaccine in children and adolescents with acute lymphoblastic leukaemia: a phase 1/2 clinical trial
Acronym/reference	TSHINE
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Sponsor	Assistance Publique – Hôpitaux de Paris
Scientific justification	Viral infections can cause serious complications in children with acute leukaemia (AL). Among these, varicella-zoster virus (VZV) infections are particularly common and can cause severe chickenpox or shingles. These infections most often occur in children with acute lymphoblastic leukaemia (ALL) during the maintenance phase of treatment. There are two live attenuated vaccines against varicella, which are contraindicated in immunocompromised individuals. No inactivated varicella vaccine is currently available. The adjuvanted recombinant zoster vaccine HZ/su (Shingrix®) is licensed for the prevention of herpes zoster in adults aged 50 years and older, and in immunocompromised individuals from 18 years of age. Studies have shown that this vaccine is well tolerated and effective in preventing herpes zoster, including in adults with hematologic malignancies. Its efficacy in preventing varicella has not yet been evaluated. Its immunogenicity in VZV-seronegative (VZV-) individuals has only been reported in a study of adult solid organ transplant recipients and appears to be lower than in VZV-seropositive individuals. In children, a phase II study conducted with a paediatric formulation of Shingrix ® (half-dose adjuvant) in VZV+ kidney transplant recipients is currently ongoing (NCT04006808), with results expected in 2027. To date, no safety signals have been reported in this study. As the paediatric formulation is not yet commercially available, some immunocompromised children have already received the adult formulation of the vaccine (full adjuvant dose) off-trial. A recently published case series of 10 immunocompromised children demonstrated that the use of this formulation elicited strong immunogenicity and was not associated with any serious adverse

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	effects. As we have no data on the safety or immunogenicity of Shingrix® in children undergoing chemotherapy and as paediatric formulation is not yet commercially available, we propose to test three dose levels of the adult vaccine in 12 patients (one half, three quarters or a full dose) in order to find the maximum dose of adjuvant tolerated in the phase 1. This chosen dose will be assessed in the phase 2 which will involve the inclusion of a further 66 participants.
Main objective and primary endpoint	**Main objectives** **Phase 1:** To define the maximum tolerated dose (MTD) of reconstituted Shingrix® vaccine for children with ALL, among three dose levels (one half, three quarters or a full dose of the adult vaccine); the MTD is defined as the highest dose level associated with a 30% probability of dose-limiting toxicity. **Phase 2**: To evaluate and compare the anti-glycoprotein E (gE) humoral immune response in children with ALL undergoing maintenance treatment and who have or have not been allocated to receive two doses of Shingrix® vaccine administered 1 month apart (dose recommended by the Phase 1), one month after second injection or two months after the randomization (M2). **Primary endpoints** **Phase 1:** Dose limiting toxicity (DLT), which is defined as any local (pain or reaction at the site of injection) or general (allergic reactions or anaphylaxis, arthralgia, myalgia, non-documented fever, headaches, gastro intestinal symptoms or Guillain Barré syndrom) treatment emerged adverse event (TEAE) grade 3 or worse occurring within 7 days following vaccine injection. ~~chills, fever, fatigue, malaise, myalgia, arthralgia, lymphadenopathy, hypersensitivity reactions including skin rash, urticaria, angioedema, headache, abdominal pain, diarrhoea, nausea and vomiting~~ **Phase 2:** Proportion of participants presenting anti-glycoprotein E (gE) humoral immune response at M2, defined as: - anti-gE IgG titer ≥ 4 x the positivity cutoff (97 mIU/mL) in VZV – subjects. - anti-gE IgG titer ≥ 4 x anti-gE IgG level measured at randomisation (M0) in VZV+ subjects. Measurement of anti-gE IgG levels will be realized by an Elisa test (kit distributed by Interchim).
Secondary objectives and endpoints	**Secondary objectives** - To assess vaccine reactogenicity within 7 days of each injection (“Vaccine” group) - To assess and compare between ~~randomised~~ groups:

Commenté [CoD5]: Ce qui est noté en page 23 faut il homogénéiser ?

Commenté [SE6R5]: Non : cela reprend les données du tableau 1 des DLT, page 26.

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	• Tolerance in the 12 months following randomisation and/or the first dose of vaccination (M0) • Anti-gE humoral immune response at M1 according to the child's VZV status • Anti-VZV humoral immune response at M0, M1, M2, according to the child's VZV status • Persistence of the humoral response (anti-VZV and anti-gE) at M6 and M12, according to the child's VZV status. • Specific T cell response at M2, according to the child's VZV status • Persistence of the specific T-cell response at M12, according to the child's VZV status. - To compare the humoral and cellular responses of vaccinated children according to their serological status regarding VZV. - To correlate the humoral and cellular vaccine responses with the immune status at the time of randomisation − To evaluate the impact of a previous treatment with Blinatumomab on immune humoral (anti-VZV and anti-gE) and cellular responses. - To describe and evaluate in an exploratory manner the rate of VZV infections or reactivations within the 12 months of follow-up in the two groups. **Secondary endpoints** - Description of the local and systemic solicited adverse reactions, any grade, within 7 days of each injection - Tolerance will be assessed by describing the occurrence of in both groups: ➢ Any grade, unsolicited adverse reactions ➢ Other adverse events: Any AEs of any grade throughout the study period including SAEs - Proportion of participants presenting anti-gE humoral immune response at M1 in the different groups and according to the Child's VZV status. [Definition of Anti-glycoprotein E humoral immune response is the same as for the primary endpoint.] - Proportion of participants presenting anti-VZV humoral immune response at M0, M1, M2 in the different groups and according to the Child's VZV status. Humoral response is defined as: • IgG anti-VZV > 165 mUI/mL in VZV - subjects at the time of inclusion. • IgG anti-VZV ≥ 4 x anti-VZV IgG level measured at M0 in VZV+ subjects at the time of inclusion. Measurement of anti-VZV IgG levels will be performed by Chemiluminescent Immunoassay using the LIAISON® VZV IgG HT kit on LIAISON® XL (Diasorin).

Commenté [CoD7]: J'ai supprimé randomisation pour inclure les particpants de la phase 1, on 'avait oublié dans le synopsis DGOS

Commenté [FAL8R7]: ok

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	- Proportion of participants presenting anti-VZV, anti-gE humoral immune response at M6 and M12, in the different groups and according to the Child's VZV status. Anti-VZV, anti-gE humoral immune response defined as previously described. - Measurement of anti-gE and anti-VZV IgG level at all time points in geometric mean titers in the different groups and according to the Child's VZV status. - Proportion of participants presenting a specific T-cell response at M2 by measurement of the number of circulating LT specific for VZV antigens (recombinant gE or VZV lysate) by double-enzymatic IFN-γ/IL-2 ELISpot technique at M0 and at M2. The evaluation of cellular immune response will be carried out using two approaches: ➢ *Ex vivo* stimulation with whole gE will allow assessment of the induction of a CD4+ T vaccine response ➢ *Ex vivo* stimulation with a lysate of VZV-infected cells will be used to assess the overall cellular response to the virus. A response is considered positive to a cytokine if the number of spots in wells stimulated with VZV antigens (recombinant gE or VZV lysate) is at least 2-fold greater than the number of spots in the negative control, using a cut-off of 10 SFC (spots forming cells) /10^5 cells after background subtraction as previously described (18). A positive response to the vaccine is defined as a 2-fold increase in cytokine-producing VZV-specific T cells at M2 compared to baseline (M0). - Proportion of participants presenting a specific T-cell response at M12 by measurement of the number of circulating LT specific for VZV antigens (recombinant gE or VZV lysate) by double-enzymatic IFN-γ/IL-2 ELISpot technique (positive response defined as previously). - Proportions of participants presenting anti-VZV, anti-gE humoral immune response at M1, M2, M6 and M12 in the different groups according to prior exposure to blinatumomab. Anti-VZV, anti-gE humoral immune response defined as previously described. - Proportions of participants presenting cellular response at M2 and M12 in the different groups and according to the Child's VZV status according to prior exposure to blinatumomab. Cellular immune response defined as previously described. - Incidence of VZV infections/reactivations within the 12 months following randomisation in the different groups. - In the event of infection/reactivation: nature and source, severity (hospitalisation, admission to intensive care unit), duration of treatment, time to resumption of chemotherapy, etc). These data will be collected from medical file.

Commenté [CoD9]: idem

Commenté [FAL10R9]: ok

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Design of the study	This is a phase 1/2 multicentre open-label clinical trial, conducted in France, with two subsequent phases. **Phase 1**: dose-finding study, which will use the Bayesian continual reassessment method (CRM): Patients will be enrolled sequentially by cohorts of 2 with escalation between cohorts based only on the DLT occurring between infusion and day 7. Dose escalation will be validated by a DLT evaluation Committee. **Phase 2**: Subjects will be randomised (ratio allocation 1:1) to 2 parallel groups: • "Vaccine" group: Participants will receive 2 doses, 1 month apart (M0 and M1), of HZ/su vaccine (Shingrix®). Vaccination will be given during maintenance therapy (at the latest during the first six months of maintenance treatment) • 'Control' group: Subjects randomised in this group will not receive either a vaccine or a Placebo injection. Randomisation will be stratified according to the child's VZV serology status at diagnosis of leukaemia (+ or -). In the event of a positive serology and a transfusion or IV immunoglobulin infusion in the month preceding the serology, it should be verified that the child has a documented history of varicella before being considered VZV-positive. In the case of a negative serology, it should be ensured that the child has not contracted a VZV infection since the start of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization.
Category	Cat 1/2: Phase 1/2
Population of study participants	Children aged 1 to 17 years with acute lymphoblastic leukaemia
Inclusion criteria	1. Children aged 1 to 17 years with acute lymphoblastic leukaemia 2. Inclusion within 6 months of the start of maintenance treatment 3. Participants with childbearing potential having reliable contraception from 4 weeks before vaccination to 12 weeks after the last vaccination, such as established use of oral, injected or implanted hormonal methods of contraception; placement of an intrauterine device (IUD) or intrauterine system (IUS); barrier methods: condom or occlusive cap (diaphragm or cervical/vault caps) with spermicidal foam/gel/film/cream/suppository; male partner sterilization (the vasectomized partner should be the sole partner for that subject); surgical sterilization (hysterectomy, bilateral oophorectomy, tubal ligation) or true abstinence (when this is in line with the preferred and usual lifestyle of the subject) prior to enrolment at M0 4. Patient affiliated to a social security scheme

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	5. Informed consent of holder(s) of legal authority and of the participant if applicable
Exclusion criteria	1. History of varicella exposure/varicella in the 21 days prior to randomisation 2. Platelet count < 50 G/L or PNN count < 0.5G/L at randomisation 3. Any bleeding disorder considered a contraindication to intramuscular injection 4. Treatment with current or prior exposure to inotuzumab 5. Vaccination against influenza within 15 days and for other vaccines within 4 weeks prior to randomisation up to M2 6. Fever within 72 hours before randomisation 7. Allergy to one of the components of the vaccine or serious allergic history (anaphylactic type) 8. History of severe adverse reactions following administration of a vaccine, including an anaphylactic reaction and associated symptoms such as rash, difficulty breathing, angioedema and abdominal pain, or a history of allergic reaction that may be exacerbated by a component of the vaccine. 9. Known infection with HIV, HCV or HBV 10. Current or planned participation in a vaccination trial 11. Participation without the investigator's agreement in other research within 4 weeks prior to the inclusion visit and for the duration of the trial.
Investigational medicinal product(s)	The investigational medicinal product of the study is HZ/su vaccine (Shingrix®) (GSK). Administration by intramuscular route will be performed at M0 and at M1 for the “vaccine” group. **Phase 1**: dose levels (DL) tested of the Shingrix® DL 1: 1/2 dose of the reconstituted Shingrix® DL 2: 3/4 dose of the reconstituted Shingrix® DL 3: full dose of the reconstituted Shingrix® **Phase 2**: recommended phase I dose
Comparator treatment	No vaccine (phase 2)
Interventions added for the study	– Shingrix® administration – 2 Teleconsultations (“Vaccine group” only) – Blood samples for ELISA and ELISpot tests – 12 Diary cards to be completed after visit M0 and M1
Expected benefits for the participants and for society	The benefits of the vaccine are based on the reduction of VZV infections, which can be serious and delay the treatment of acute leukaemia, regardless of the severity of the infection, thereby increasing the risk of relapse.
Risks and burdens added by the study	- Local and general reactions due to the vaccine - Risks associated with venous sampling include haematoma at the puncture site, pain at the puncture site, malaise, mild bleeding and anxiety.

Commenté [LGC11]: ajoute-t-on femmes enceintes ou allaitantes ?
critère purement réglementaire

Commenté [FAL12R11]: honnetement il n'y a pas de femmes enceintes ou allaitantes en cours de traitement d'entretien de LAL...

Commenté [CoD13]: Car pas de comparateur dans la phase 1

Commenté [FAL14R13]: ok

Commenté [PC15]: Préciser une pour chaque phase

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Practical implementation	In this prospective multicentre, randomised controlled open-label clinical trial, we will include children aged 1 to 17 years with acute lymphoblastic leukaemia. In phase 2 of trial, they will be randomly assigned to Shingrix® vaccine or no vaccine.
Number of participants included	Total: 78 **Phase 1:** 12 (VZV- and VZV +) **Phase 2:** 66 (40 VZV- and 26 VZV+)
Number of centres	6 centres in France, specialised in Oncology, Haematology **Phase 1**: 1 centre *If recruitment 'is slow', we may involve other APHP centres. All centres that have confirmed their participation will take part in Phase II.* **Phase 2**: 6 centres
Duration of the study	- *Inclusion period:* 30 months - *Participation period (treatment + follow-up):* 12 months - *Total duration :* 42 months
Number of enrolments expected per site and per month	0.7 to 2.1
Statistical analysis	**Phase 1**: A Bayesian framework will be used to determine the probability of DLT (primary outcome). Sequential analyses will be performed every 2 enrolled patients. **Phase 2**: Given the uncertainty in estimated parameters of the sample size computation, an interim analysis will be conducted with re-evaluation of sample size after the inclusion of 33 patients (that is, one half of the scheduled sample size). Terminal analysis will be based on an intent-to-vaccine principle, once the patients have been enrolled, stratified on the stratum of randomisation (i.e., the child's VZV serology status at diagnosis).
Funding sources	Ministry of Health
Study will have a Data Safety Monitoring Board	Yes. This study is at D risk.

Commenté [CoD16]: Faut il preciser pour la phase 2 uniquement

Commenté [SE17R16]: Précision apportée

Commenté [CoD18]: Cela me semble important de le preciser (info notée p51)

Commenté [FAL19R18]: ok

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2. SCIENTIFIC JUSTIFICATION FOR THE STUDY

2.1. Context of the study

Viral infections can cause serious complications in children with acute leukaemia (AL) (1). Among these, VZV infections are particularly common (2, 3) and can cause severe chickenpox or shingles (4-7). These infections most often occur during treatment but can also occur in the months following discontinuation (8).

According to studies, the incidence of herpes zoster in paediatric patients ranges from 42 to 262 per 100,000 person-years (9). In a cohort study conducted in Taiwan, the relative risk of developing herpes zoster was 8.6 times higher (95% CI: 4.8–15.6) in children with cancer, with acute leukaemia being the cancer most strongly associated with this risk (2). Moreover, the occurrence of varicella or herpes zoster can delay chemotherapy, exposing these patients to a higher risk of relapse.

Despite current therapies, which are based on the early initiation of antiviral treatment and the administration of specific intravenous immunoglobulins as part of post-exposure prophylaxis, the occurrence of varicella or herpes zoster remains associated with high morbidity and mortality in children with acute leukaemia (AL). These patients are highly immunocompromised and therefore at greater risk of severe disseminated varicella, which can lead to intensive care unit admission and, in some cases, death, or complicated herpes zoster. Furthermore, the occurrence of these infections necessitates the temporary interruption of chemotherapy, which may negatively impact leukaemia prognosis (2, 10).

In a recent study on the characteristics of varicella-zoster virus (VZV) infections in children with malignant hematologic diseases, infections were found to occur more frequently: 1) in children with acute lymphoblastic leukaemia (ALL) and 2) during the maintenance phase of treatment, when children return to group settings (8).

These findings highlight the need for preventive strategies to reduce morbidity and mortality associated with these infections in this vulnerable population.

Two vaccines have been approved for paediatric use against chickenpox in France. Both are live attenuated vaccines and are therefore contraindicated during chemotherapy treatment (11). Currently no inactivated varicella vaccine is available. Additionally, no paediatric herpes zoster vaccine has received marketing authorization in France.

Based on promising safety, immunogenicity and efficacy data in adults at risk of for herpes zoster, the adjuvanted recombinant HZ/su Shingrix® vaccine was granted marketing authorisation for the prevention of herpes zoster and postherpetic neuralgia in adults aged 50 and over on 21 March 2018, and in immunocompromised adults aged 18 and older on 25 August 2023. This vaccine has been available in France since 2 October 2023 and is reimbursed since 14 December 2024 for people aged 65 and over and those aged 18 and over with an increased risk of shingles.

Because >95% of adults are immunized against VZV, ~~Its~~ ability of HZ/su Shingrix® vaccine to prevent the onset of varicella, ~~which is infantile disease (median age of varicella is 4 years),~~ has not yet been studied. However, as this vaccine targets VZV, it could also prevent chickenpox in VZV-seronegative patients, which would be particularly beneficial in pediatric populations, given that more than 90% of varicella cases

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occur in children aged 1 to 9 years. ~~–~~Its immunogenicity in VZV-negative subjects (VZV-) has only been reported in a study of adult solid organ transplant recipients, and appears to be less effective than in VZV+ subjects (12).

In children, a phase II study conducted with a paediatric formulation of Shingrix® (half-dose adjuvant) in VZV+ kidney transplant recipients is currently ongoing (NCT04006808), with results expected in 2027. To date, no safety signals have been reported in this study.

As the paediatric formulation is not yet commercially available, some immunocompromised children have already received the adult formulation of the vaccine (full adjuvant dose) off-trial. A recently published case series of 10VZV+ immunocompromised children demonstrated that the use of this formulation elicited strong immunogenicity and was not associated with any serious adverse effects (13).

As the paediatric formulation of GSK's vaccine will not be commercially available before 2032, as the adult vaccine cannot be reconstituted at half the adjuvant dose (risk of glycoprotein precipitation), and as there is an urgent need to protect children with acute leukaemia against VZV we therefore aim to conduct a phase I/II randomised controlled open-label trial designed to evaluate the safety and immunogenicity of the HZ/su Shingrix® vaccine in children with ALL to develop vaccine recommendations for this population.

This pioneering study addresses a significant public health challenge for children with acute leukaemia. It aims to generate new knowledge on the safety and immunogenicity of this vaccine in this specific population, providing valuable insights for the national and international medical community to optimize vaccination strategies.

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2.2. Summary of relevant pre-clinical experiments and clinical trials

A total of five clinical trials evaluated humoral and CD4 [+2] lymphocyte (CD4 T cells expressing ≥2 of four assessed activation markers) immune responses to Shingrix® vaccine in immunocompromised adults over 18 years of age. The majority of studies followed participants for 12 months after administration of Shingrix® HZ/su vaccine. The main results concerning the immune responses induced by the administration of two doses of the vaccine, with the exception of the clinical trial in persons who are living with HIV, were summarised in the HAS report of February 2024 (14) and are presented below:

In adults aged 18 and over who have received a Hematopoietic Stem Cell (HSC) transplant

Two studies have evaluated the immune responses induced by Shingrix® vaccine in adults after HSC transplantation. Firstly, Stadtmauer et al. (15) performed a post hoc analysis of humoral-mediated and CD4[+2] T lymphocyte immune responses according to age and initial pathology. The ATP cohort included 127 adults who received a first dose of vaccine or placebo 50 to 70 days after autologous HSC transplantation, and a second dose one or two months later. A total of 69 participants (39 vaccinated and 30 placebo) were included in the humoral

Commenté [CPE20]: Suite à la question lors de la réunion GIO et la demande dans le CR : *« il est demandé au promoteur de bien étayer le rationnel concernant l'inclusion d'enfants VZV- dans le protocole de l'essai et les attentes dans cette sous population lors de la soumission du protocole »*
⇨Sauf si plus décrit par ailleurs dans le protocole, il me semble qu'il serait important de développer davantage ici au risque d'avoir la question en RFI

@fanny : est-ce que tu penses que ça peut être intéressant, pour montrer l'intérêt d'inclure des patients VZV-, (si c'est le cas bien sûr), d'expliquer qu'une majorité des patients que tu penses inclure seront dans la tranche basse des âges, avec plus de chance d'avoir un profil VZV- ?
Je pense que c'est important de donner tous les arguments qu'on peut avoir pour montrer que tous les status VZV sont importants dans cette étude (particulièrement VZV- pour lesquels il n'y a aucune alternative et pas de développement en cours), même si on n'a pas ou peu de données chez VZV- avec ce produit.
Est-ce qu'on aurait des données d'épidémio ou autres dans la LAL avec des tranches d'âges pouvant nous permettre d'extrapoler ?

Commenté [CP21R20]: + de 94% des cas de varicelles surviennent chez les enfants de 1 à 9 cas donc notre population ont aura VZV – et VZV + (à confirmer par Fanny)

Commenté [FAL22R20]: J'ai complété pour justifier

a mis en forme : Police :(Par défaut) Times New Roman, 12 pt, Français (France)

Commenté [CPE23]: Est-ce qu'on dispose dans la publi du statut VZV ? Si oui, ajout des précisions par rapport au statut pourrait être interessant, non ? et sinon, je préciserais de façon succinte qu'on ne dispose pas du statut VZV dans la publication.

Commenté [CPE24]: Attention, le PIP de GSK n'est que dans l'indication Zona, pas dans la varicelle donc à l'AMM, seuls les patients VZV- feront partis de la pop cible.

Commenté [CPE25]: En relisant, et suite à la réunion avec le GIO, je me pose des questions notamment sur l'indication cible de ces essais qui n'est pas vraiment mis en exergue ici (Zona très certainement ? car pop adulte donc patients VZV+ ?)... je sais qu'on ne veut pas mettre en exergue ce point, mais si, dans le statut est renseigné dans la publi et que quelques adultes VZV- ont été inclus, ca pourrait appuyer notre demande d'inclure les patients VZV- ? et à défaut de statut renseigné dans la publi, je le préciserai ici par une mention... vraiment navrée pour mes questions additionnelles
En fait je me dis que notre transparence peut être appréciée sur ce point qui peut faire débat et faciliter l'évaluation du projet ensuite par l'ANSM (mais ce n'est que mon avis 😊)

Commenté [CP26R25]: A vérifier

Commenté [FAL27R25]: Le statut VZV n'est pas renseigné dans les études mais la plupart ont des anticorps anti GE avant vaccination.
Je pense qu'il faut laisser comme ca et on verra si on a des question

Commenté [CoD28]: Faut il ajouter la signification de [2+] expliqué dans l'article : (CD4 T cells expressing ≥2 of four assessed activation markers) car pas forcement clair dau premier abord

Commenté [SE29R28]: @Fanny : Qu'en pensez-vous ?

Commenté [FAL30R28]: ok je l'ai ajoutée

a mis en forme : Police :(Par défaut) Arial, 11 pt, Couleur de police : Automatique, Anglais (Royaume-Uni)

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immunogenicity analysis. Vaccine responses one month after the second dose were higher in adults aged 50 and over than in adults aged 18-49 (71.4% versus 57.7%).

Two years after vaccination, vaccine responses decreased in both age groups (VTRs fell to 23.1% and 56% in adults aged 50 and over and adults aged 18-49). An increase in anti-gE GMTs was also observed in both age groups, one month after administration of the two doses. In younger adults, the values were 12 times higher than the pre-vaccination values, and in adults aged 50 and over, they were more than 19 times higher than the pre-vaccination values. Two years after vaccination, a reduction of approximately 88% and 69% in anti-gE antibody titres was observed in younger adults and adults aged 50 and over (MGT: 1,492.5 and 4,025 respectively).

Commenté [CPE31]: => abréviation à ajouter dans le glossaire ?

Commenté [CPE32]: => abréviation à ajouter dans le glossaire ?

One month after the second dose, younger adults had a higher CD4 TRV [+2] than adults aged 50 and over (100% versus 89.3%). Vaccine responses remained at 100% two years after vaccination in the younger adults, while this rate fell to 58.8% in the older participants. In addition, median CD4 [+2] cell counts were higher, particularly in the younger participants than in those aged 50 and over (12,365.5 versus 3,294.2). However, two years after administration of the two doses, a 72% reduction in median CD4 [+2] cell counts was observed in the younger participants, while the rate of reduction was 54% in adults aged 50 and over (CD4 [+2] cells: 3,466 and 1,519.6).

Commenté [CPE33]: => abréviation à ajouter dans le glossaire ?

The study by Bastidas et al. (16) included the analysis of anti-gE antibody concentrations as a secondary endpoint in HSC transplant patients. Anti-gE TRV reached a peak of 67.1% one month after vaccination. This rate then decreased to 44.7% two years after vaccination. MGT peaked at two months post-vaccination (12,753.2 [7,973; 20,399.4]). However, a decline in antibody titres was observed two years after Shingrix® administration. Moreover, the CD4 TRV [+2] was higher after the second dose of the vaccine (93%) in the 41 vaccinated patients. Median CD4 [+2] titres peaked within two months of vaccination. However, they declined from twelve months post-vaccination, although they remained above pre-vaccination levels.

In adults aged 18 and over with haematological malignancies

Dagnew et al. (17) conducted a phase III clinical trial to assess humoral and cellular immunogenicity in adults aged 18 and over with haematological malignancies (HM). To do this, the authors included 415 participants to assess humoral immune responses and 132 participants to assess CD4 [+2] lymphocyte responses. Of the participants who received the Shingrix® vaccine, 36% were taking immunosuppressive treatment at the time of vaccination, and 64% were vaccinated after completing this treatment.

One month after Shingrix® administration, the anti-gE TRV was 65.4%, and this fell to 52.1% one year after vaccination. A 14-fold increase in anti-gE antibody titres was observed one month after the second dose (964 versus 13,445.6). In addition, the vaccinated group showed a statistically significant increase in anti-gE antibody titres, one month after vaccination, compared with the placebo group (MGC ratio: 29.75 [21.09; 41.96], $p < 0.0001$). However, a decline in anti-gE MGT of around 60% (52,022.7 to 13,445.6) was observed one year after vaccination.

Commenté [CPE34]: => abréviation à ajouter dans le glossaire ?

Commenté [CPE35]: => abréviation à ajouter dans le glossaire ?

CD4 [+2] vaccine responses increased within two months of vaccination (83.7%), and decreased by approximately 15% one-year post-vaccination (66.7%). Vaccinated participants

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also experienced a fairly significant increase in CD4 [+2], with median titres rising from 77.5 to 3,081.9 mIU/mL. However, they fell one year after receiving the two doses (1,006.7 mIU/mL), although they remained higher than the pre-vaccination values.

In adults aged 18 and over with solid tumours

In 2019, Vink et al. (18) conducted a phase II/III clinical trial to evaluate immune responses in adults with solid tumours. Patients received two doses of Shingrix® (n = 87) or two doses of placebo (n = 108), one or two months apart.

At one-month post-vaccination, there was a statistically significant increase in anti-gE antibodies in the vaccine group compared to the placebo group (MGC ratio: 14.4 [10.7; 19.5] p<0.0001). A more pronounced increase was observed in patients who had not yet started chemotherapy (MGC ratio = 23.2, p < 0.001). Vaccinated patients reached peak antibody titres one month after receiving the first dose of vaccine. Antibody titres decreased slightly after the second dose of Shingrix®. The percentage of responders reached a peak of 86.2%, which remained similar after the second dose. TRV decreased from six months onwards, with a more pronounced reduction one year after vaccine administration (51.5%).

Vaccinated patients (n=27) showed a statistically significant approximately 10-fold increase in CD4 [+2] titres compared with the placebo group (MGC ratio: 9.94 [3.63; 27.19], p<0.0001). Nevertheless, two months after vaccine administration, the CD4 TRV [+2] was 50% (95% CI: 28.2; 71.8), and according to the evaluation criteria stipulated in the study protocol, the TRV did not reach the established end-point (threshold of the lower limit of the 95% CI ≥ 50%).

2.3. Description of the population to be studied and justification for this choice of participants

Severe forms of chickenpox and shingles occur in children with acute lymphoblastic leukemia (ALL), mainly during maintenance therapy. Moreover, these infections can delay chemotherapy, potentially increasing the risk of relapse.

These findings highlight the need for preventive strategies to reduce morbidity and mortality associated with these infections in this vulnerable population, in which lived vaccines are contraindicated.

Studies have shown that the adjuvanted recombinant HZ/su Shingrix® vaccine is well tolerated and effective in preventing shingles in immunocompromised adults. However, its ability to prevent chickenpox has not been studied.

No clinical studies have been published in immunocompromised children, but a trial is currently underway in kidney-transplanted children aged 1 to 17 years to assess the tolerance and immunogenicity of the vaccine in the herpes zoster indication (2 doses 1 month apart, starting 6 months after transplantation). For this study, GSK gave a paediatric formulation containing the same dose of antigen as that used in the Shingrix® 'adult' vaccine, but with a dose of adjuvant divided by 2. As the paediatric formulation is not yet commercially available, some immunocompromised children have already received the adult formulation of the vaccine (full adjuvant dose) off-trial. A recently published case series of 10 immunocompromised children

Commenté [CPE36]: Il faut ajouter à mon sens que l'étude est dans la prévention du zona.
J'ajouterais aussi le numéro de l'essai pour qu'il soit plus facilement retrouvé : **EudraCT Number:** 2019-000607-33

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demonstrated that the use of this formulation elicited strong immunogenicity and was not associated with any serious adverse effects (13).

As the paediatric formulation of GSK's vaccine will not be commercially available before 2032, as the adult vaccine cannot be reconstituted at half the adjuvant dose (risk of glycoprotein precipitation), and as there is an urgent need to protect children with acute leukaemia against VZV, we therefore aim to conduct a phase I/II randomised controlled open-label trial designed to evaluate the safety and immunogenicity of the HZ/su Shingrix® vaccine in children with ALL to develop vaccine recommendations for this population.

Commenté [CPE37]: Et dans l'indication zona… pas dans l'indication varicelle

Commenté [CPE38]: Et je me demande si ca ne vaudrait pas le coup d'appuyer qu'il n'y a pas d'étude en cours pour proposer une formulation pédiatrique pour la prévention de la varicelle chez l'enfant immunodéprimé (donc qu'il y a un vrai besoin médical chez les patients VZV- également d'où l'inclusion également de patient VZV- malgré l'absence de données d'efficacité de ce vaccin dans cette indication).

2.4. *Hypotheses for the study*

1- Vaccination of children aged 1 to 17 years with lymphoblastic acute leukaemia (ALL) during
1- Vaccination of children aged 1 to 17 years with lymphoblastic acute leukaemia (ALL) during
1- Vaccination of children aged 1 to 17 years with lymphoblastic acute leukaemia (ALL) during

Commenté [CP39]: A supprimer et mettre dans les objectifs

Commenté [FAL40R39]: fait

~~1~~12~~-~~ Vaccination of children aged 1 to 17 years with lymphoblastic acute leukaemia (ALL) during maintenance treatment with HZ/su Shingrix® vaccine (two doses (dosage of each dose defined during phase 1) of vaccine, administered 1 month apart) will be well tolerated.

Commenté [CoD41]: Je pense qu'on ne peut plus mettre cette phrase en l'etat vu que nous allons tester plusierus dosages

Commenté [SE42R41]: Ajouter un point concernant la phase 1 ?

Commenté [FAL43R41]: J'ai changé la phrase, est ce que ca vous va ?

2- The immunogenicity of the group of children with ALL who receive Shingrix® HZ/su vaccine will be significantly superior to that of the group of children with ALL who do not receive the vaccine.

3- The number of cases of chickenpox and zoster and the number of days of delay in chemotherapy treatment will be reduced in the group of children with acute leukaemia who receive the HZ/su Shingrix® vaccine.

2.5. *Identification and description of the investigational medication*

We plan to use the commercially available adjuvanted recombinant HZ/su vaccine (Shingrix®) in the study. See Section 7 for further information about the product.

2.6. *Description and justification of the dosage, route of administration, administration schedule and treatment duration*

Commenté [SE44]: @ AGEPS : à vérifier

The current recommended administration schedule for Shingrix® vaccine in immunocompromised adults is as follows: 2 doses of 0.5 mL (50µg) administered 2 months apart. The interval may be reduced to 1 month or extended to 6 months if justified by disease or treatment.

a mis en forme : Police :Symbol

The vaccine is marketed in the following form:

- 1 vial of powder with a removable brown cap (antigen)
- 1 vial of suspension with a removable blue-green cap (adjuvant), for one dose = 0.5 mL

After reconstitution, one dose (0.5 mL) contains:

- Varicella-Zoster Virus glycoprotein E antigen: 50 micrograms
- Adjuvant AS01B containing:

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- Quillaja saponaria Molina plant extract, fraction 21 (QS-21): 50 micrograms
- 3-O-desacyl-4'-monophosphoryl lipid A (MPL) from Salmonella minnesota: 50 micrograms

During the study, each patient will receive a regimen consisting of 2 intramuscular injections of Shingrix® HZ/su vaccine 1 month apart and the first dose will be given within 6 months of the start of maintenance treatment in children with ALL.

As we have no data on the safety or immunogenicity of Shingrix® in children undergoing chemotherapy and as paediatric formulation is not yet commercially available, we propose to test three dose levels of the adult vaccine in 12 patients (one half, three quarters or a full dose) in order to find the maximum dose of adjuvant tolerated in the phase 1 of the trial before starting the phase 2 of trial.

As we have no data on the safety or immunogenicity of Shingrix® in children undergoing chemotherapy and as paediatric formulation is not yet commercially available,~~,~~ we propose to test three dose levels of the adult vaccine in 12 patients (one half, three quarters or a full dose) in order to find the maximum dose of adjuvant tolerated in the phase 1 of the trial before starting the phase 2 of trial.

See Section 7 for further information about the administration of the product.

2.7. Summary of the known and foreseeable benefits and risks for the Clinical Trial participants

The benefits of the vaccine are based on the reduction of VZV infections, which can be serious and delay the treatment of acute leukaemia, regardless of the severity of the infection, thereby increasing the risk of relapse.

The risks associated with the vaccine and procedures are as follows:

- Local reaction: pain, swelling, redness, pruritus at the injection site.
- General reaction: chills, fever, fatigue, malaise, myalgia, arthralgia, lymphadenopathy, hypersensitivity reactions including skin rash, urticaria, angioedema, headache, abdominal pain, diarrhoea, nausea and vomiting or Guillain Barré syndrome.
- Risks associated with venipuncture and venous sampling include haematoma at the puncture site, pain at the puncture site, malaise, mild bleeding and anxiety.

3. OBJECTIVES

3.1. Primary objectives

Phase 1: To define the maximum tolerated dose (MTD) of reconstitued Shingrix® vaccine for children with ALL, among three dose levels of the adult vaccine (one half, three quarters or a full dose); the MTD is defined as the highest dose level associated with a 30% probability of dose-limiting toxicity.

Phase 2: To evaluate and compare the anti-glycoprotein E (gE) humoral immune response in children with ALL undergoing maintenance treatment and who have or have not been allocated

Commenté [CoD45]: Je ne le mettrai pas la phase 1 est surtout pour la safety notamment au vu de l'effectif

Commenté [CoD45]: Je ne le mettrai pas la phase 1 est surtout pour la safety notamment au vu de l'effectif

Commenté [SE46R45]: @AGEPS ?

Commenté [SE46R45]: @AGEPS ?

Commenté [FAL47R45]: on a mis en objectif que c'était pour tester la dose max tolérée non ?

Commenté [FAL47R45]: on a mis en objectif que c'était pour tester la dose max tolérée non ?

Commenté [CPE48R45]: A voir plutôt avec la PV je pense, mais en effet la phase I est la recherche de dose et donc de safety… L'immunogénicité n'est évaluée que dans la deuxième phase sur la dose trouvée en phase 1

Commenté [CPE48R45]: A voir plutôt avec la PV je pense, mais en effet la phase I est la recherche de dose et donc de safety… L'immunogénicité n'est évaluée que dans la deuxième phase sur la dose trouvée en phase 1

Commenté [LGC49R45]: Tout à fait
Le paragraphe proposé est ok pour moi

Commenté [LGC49R45]: Tout à fait
Le paragraphe proposé est ok pour moi

Commenté [FAL50]: @AGEPS est ce que vous justifieriez plus ?

Commenté [FAL50]: @AGEPS est ce que vous justifieriez plus ?

Commenté [CJ51R50]: C'est suffisant ainsi pour moi

Commenté [CJ51R50]: C'est suffisant ainsi pour moi

Commenté [CPE52R50]: Ok aussi pour moi

Commenté [CPE52R50]: Ok aussi pour moi

Commenté [SE53]: @ Vigilance

Commenté [LGC54R53]: C'est ok pour moi

Commenté [CoD55]: On ne mets le syndrome de Guillain Barré comme mis dans les DLT dans le resumé

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to receive two doses of Shingrix® vaccine administered 1 month apart (dose recommended by the Phase 1), one month after the second injection or two months after randomization (M2).

3.2. Secondary objectives

- To assess vaccine reactogenicity within 7 days of each injection (“Vaccine” group).
- To assess and compare between groups:
  - Tolerance in the 12 months following randomisation and/or the first dose of vaccination (M0).
  - Anti-gE humoral immune response at M1, according to the child's VZV status.
  - Anti-VZV humoral immune response at M0, M1, M2, according to the child's VZV status.
  - Persistence of the humoral response (VZV and gE) at M6 and M12, according to the child's VZV status.
  - Specific T cell response at M2, according to the child's VZV status.
  - Persistence of the specific T-cell response at M12, according to the child's VZV status.
- To compare the humoral and cellular responses of vaccinated children according to their serological status regarding VZV.
- To correlate the humoral and cellular vaccine responses with the immune status.
- To evaluate the impact of a previous treatment with Blinatumomab on immune humoral (anti-VZV and anti-gE) and cellular responses.
- To describe and evaluate in an exploratory manner the rate of VZV infections/reactivations within the 12 months of follow-up in the different groups.

4. STUDY DESIGN

4.1. Study design

This is a phase 1/2 clinical trial conducted in children with ALL.

Phase 1: Continual Reassessment Method (CRM) design
This is an adaptive sequential method initially proposed for dose-finding studies.
It is close to the standard 3+3 design

- Here, several dose levels 1/2, 3/4, and full dose of the reconstituted Shingrix® vaccine, will be available for testing.
- It uses a sequential enrolment of patients by cohorts treated at the same dose level.
- It is ethically based: enrolment begins at the less exposed dose level (for the patient).
- It is adaptive: it uses the data from the previous cohorts to modify the dose level for further cohorts.

However, there are main differences that outperforms the CRM over the standard 3+3 design:

- It aims at defining a dose level associated with a targeted probability of DLT whereas 3+3 design’s target is not formally defined.
- It can use cohorts of various sizes (not necessarily of 3, here of 2) at each dose level.

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- It uses an underlying mathematical model relating dose level and probability of DLT, that allows an estimation procedure – whereas 3+3 only provides an optimal dose level without any estimate of probability on the sole basis of empiric observation.
- It uses all information recorded in the trial – while 3+3 only uses the last 3 (or 6) observations.

Phase 2

This is a multicenter randomised (ratio allocation 1:1) design, with 2 parallel groups:

- "Vaccine" group: Participants will receive 2 doses, 1 month apart (M0 and M1), of HZ/su vaccine (Shingrix®), at the recommended dose following the phase 1. Vaccination will be given during maintenance therapy (at the latest during the first six months of maintenance treatment)
- 'Control' group: Subjects randomised in this group will not receive either a vaccine or a Placebo injection.

Randomisation will be stratified according to the child's VZV serology status(+ or -) of leukaemia .In the event of a positive serology and a transfusion or IV immunoglobulin infusion in the month preceding the serology, it should be verified that the child has a documented history of varicella before being considered VZV-positive. In the case of a negative serology, it should be ensured that the child has not contracted a VZV infection since the start of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization.

Commenté [CPE56]: Je ne comprends plus la phrase reformulée car c'est le statut de la sérologie VZV et non le statut de la leucémie qui est + ou - => la phrase en anglais n'a plus le sens qu'on recherchait en français je pense. Je préférais la formulation initiale. Proposition « at the time of the maintenance phase of LAL »

Commenté [SE57]: A supprimer pour homogénéiser page 25 ?

Commenté [CPE58]: Je ne comprends plus la phrase reformulée car c'est le statut de la sérologie VZV et non le statut de la leucémie qui est + ou - => la phrase en anglais n'a plus le sens qu'on recherchait en français je pense. Je préférais la formulation initiale. Proposition « at the time of the maintenance phase of LAL »

4.2. Study endpoints

4.1.1. Primary endpoints

Phase 1: Dose limiting toxicity (DLT), which is defined as any local (pain or reaction at the site of injection) or general (allergic reactions or anaphylaxis, arthralgia, myalgia, non-documented fever, headaches, gastro intestinal symptoms or Guilllain Barré Syndrome) treatment emerged adverse event (TEAE) grade 3 or worse occurring within 7 days following vaccine injection (table 1).

				DLT	
		Mild (Grade 1)	Moderate (Grade 2)	**Severe (Grade 3)**	**Very severe (Grade 4)**
LOCAL REACTIONS	**Pain at the injection site**	Does not interfere with activities	Requires the use of a level 1 or 2 analgesic for more than 24 hours	**Requires the use of a level 3 analgesic**	**Requires hospitalization**
	Injection-site tissue reaction	Erythema or induration < 2 cm	Erythema or induration 2.1–5 cm; lymphadenopathy	**Necrosis or ulceration**	**Severe tissue damage requiring surgical intervention**

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GENERAL REACTIONS	**Allergic reaction or anaphylaxis**	Skin reaction not requiring treatment	Skin reaction requiring treatment	**Symptomatic bronchospasm, with or without urticaria; allergic edema/angioedema; hypotension**	**Life-threatening condition requiring intensive care management**
	Arthralgia, myalgia	Does not interfere with activities	Requires the use of a level 1 or 2 analgesic for more than 24 hours	**Requires the use of a level 3 analgesic**	**Requires hospitalization**
	Gastrointestinal symptoms	Does not interfere with activities	Requires symptomatic treatment for more than 24 hours	**Requires hospitalization**	—
	Fever without documented infectious cause	≥ 38.0°C not requiring oral treatment	≥ 38.0°C requiring oral treatment	**≥ 38.0°C requiring hospitalization**	**≥ 38.0°C, life-threatening, requiring intensive care management**
	Headache not related to ~~post-lumbar puncture CSF leak~~	Requires a level 1 or 2 analgesic for less than 24 hours	Requires a level 1 or 2 analgesic for more than 24 hours	**Requires a level 3 analgesic**	**Requires hospitalization**
	Guillain–Barré syndrome	—	—	**Presence of clinical involvement**	**Life-threatening condition requiring intensive care management**

Table 1. Definitions of DLTs related to local or systemic reactions secondary to vaccination.

Commenté [CoD59]: Remettre les titres quand le tableau sera validé

Commenté [SE60R59]: A compléter

Phase 2: Proportion of participants presenting anti-gE humoral immune response at M2 in the different groups, defined as:

- anti-gE IgG titer ≥ 4 x the positivity cutoff (97 mIU/mL) in VZV - subjects.
- anti-gE IgG titer ≥ 4 x anti-gE IgG level measured at randomisation (M0) in VZV+ subjects.

Measurement of anti-gE IgG levels will be realized by an Elisa test (kit distributed by Interchim).

4.1.2. Secondary endpoints

- Description of the local and systemic solicited adverse reactions, any grade, within 7 days of each injection
- Tolerance will be assessed by describing the occurrence of in both groups:
  - Any grade, unsolicited adverse reactions
  - Other adverse events: Any AEs of any grade throughout the study period including SAEs

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- Proportion of participants presenting anti-gE humoral immune response at M1 in the different groups and according to the Child's VZV status. [Definition of Anti-glycoprotein E humoral immune response is the same as for the primary endpoint.]
- Proportion of participants presenting anti-VZV humoral immune response at M0, M1, M2 in the different groups and according to the Child's VZV status. Immune response is defined as:
  - IgG anti-VZV > 165 mUI/mL in VZV - subjects at the time of inclusion.
  - IgG anti-VZV ≥ 4 x anti-VZV IgG level measured M0 in VZV+ subjects at the time of inclusion.

  Measurement of anti-VZV IgG levels will be performed by Chemiluminescent Immunoassay using the LIAISON® VZV IgG HT kit on LIAISON® XL (Diasorin).
- Proportion of participants presenting anti-VZV, anti-gE humoral immune response at M6 and M12, in the different groups and according to the Child's VZV status. Anti-VZV, anti-gE humoral immune responses are defined as previously described.
- Measurement of anti-gE and anti-VZV IgG level at all time points in geometric mean titers in the different groups and according to the Child's VZV status.
- Proportion of participants presenting a specific T-cell response at M2 by measurement of the number of circulating LT specific for VZV antigens (recombinant gE or VZV lysate) by double-enzymatic IFN-γ/IL-2 ELISpot technique at M0, and M2.

The evaluation of cellular immune response will be carried out using two approaches:

- *Ex vivo* stimulation with whole gE will allow assessment of the induction of a CD4+ T vaccine response
- *Ex vivo* stimulation with a lysate of VZV-infected cells will be used to assess the overall cellular response to the virus.

A response is considered positive to a cytokine if the number of spots in wells stimulated with VZV antigens (recombinant gE or VZV lysate) is at least 2-fold greater than the number of spots in the negative control, using a cut-off of 10 SFC (spots forming cells) /10^5 cells after background subtraction as previously described (19). A positive response to the vaccine is defined as a 2-fold increase in cytokine-producing VZV-specific T cells at M2 compared to baseline (M0).

- Proportion of participants presenting a specific T-cell response at M12 by measurement of the number of circulating LT specific for VZV antigens (recombinant gE or VZV lysate) by double-enzymatic IFN-γ/IL-2 ELISpot technique (positive response defined as previously).
- Proportions of participants presenting anti-VZV, anti-gE humoral immune response at M1, M2, M6 and M12 in the different groups according to prior exposure to blinatumomab. Anti-VZV, anti-gE humoral immune response defined as previously described.
- Proportions of participants presenting cellular response at M2 and M12 in the different groups and according to the Child's VZV status according to prior exposure to blinatumomab. Cellular immune response defined as previously described.
- Incidence of VZV infections/reactivations within the 12 months following randomisation in the different groups.

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- In the event of infection/reactivation: nature and source, severity (hospitalisation, admission to intensive care unit), duration of treatment, time to resumption of chemotherapy, etc). These data will be collected from medical file.

4.2. Description of research methodology

4.2.1. Conduct of the study

After the persons with parental authority have signed an informed consent form, participants will undergo an inclusion medical visit including a clinical examination and a blood test to check their eligibility.

In Phase 1, the CRM design will be applied as follows:

- Patients will be enrolled sequentially, by cohorts of 2 patients.
- The first cohort will receive the first dose level.
- Note that, for ethical and safety concerns, inclusion of the first patient at a new dose will be delayed until evaluation at day 7 following the first injection of vaccine of the first patient from the previous cohort.
- Once a cohort has been completed, based on the observed endpoint (i.e., occurrence or not of DLT within the first 7 days post each infusion of vaccine) of the 2 patients from this cohort, the estimation procedure is rerun, and recommended dose level for patients in the second cohort will use observed responses (DLT or not) of each previously enrolled patient.
- Note that for ethical concerns, no skipping will be allowed so that no dose level could be recommended unless all previous levels have been tested. The process is continued until the sample size of 12 patients has been reached.

In Phase 2, participants will then be randomised (ratio allocation 1:1) to two parallel groups, either the “Vaccine” or “Control” group. Randomisation will be stratified according to the child's VZV serology status at diagnosis of leukaemia (+ or -). If serology is positive and the child has received a transfusion or intravenous immunoglobulin (IVIg) infusion within the preceding month, a documented history of varicella must be confirmed before considering the child as VZV-positive. If serology is negative, it must be verified that the child has not developed a VZV infection since the initiation of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization.

- “Vaccine” group:

Participants will receive 2 doses, 1 month apart (M0 and M1), of HZ/su vaccine (Shingrix®).

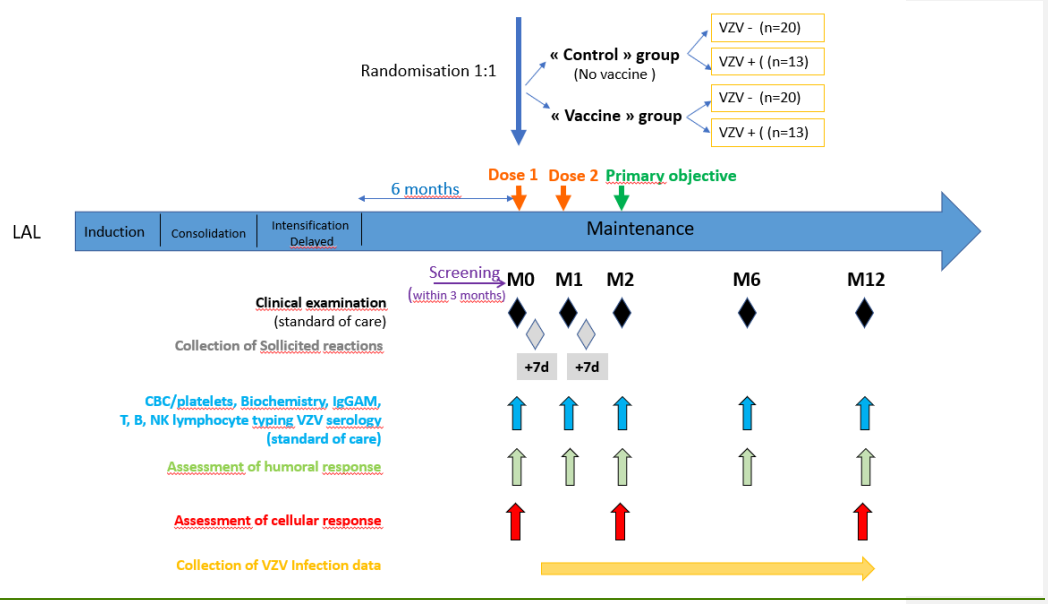
Vaccination will be given during maintenance therapy (at the latest during the first six months of maintenance treatment).

- “Control” group

Subjects randomised in this group will not receive vaccine.

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Summary diagram of the phase 2 TSHINE study.



4.2.2. *Number of participating sites*

This is a multicentre national clinical trial involving 6 centres in France (cf. addendum 1). The participants will be recruited among outpatients during a hospital visit with the specialists. For Phase I, the Trousseau centre will take part; if recruitment 'is slow', we may involve other APHP centres. All centres that have confirmed their participation will take part in Phase II.

4.2.3. *Identification of participants*

The participants in this Clinical Trial will be identified as follows:
Site number (3 digits) - Sequential enrolment number for the site (4 digits)

This reference number is unique and will be used for the entire duration of the study.

A randomisation number will also be assigned when the participant is randomised. This number will have the following format: RXXX.

Commenté [SE62]: Question de Christine : Il n'y aura pas de numéro de traitement ?

Commenté [SE63R62]: @AGEPS : A vérifier avec AGEPS mais pas nécessaire car essai en ouvert, seul le vaccin est dispensé

Commenté [CJ64R62]: Essai en ouvert donc pas de numéro de traitement.
Si l'URC n'utilise pas ce numéro de randomisation, nous pouvons le supprimer

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4.2.4. Randomisation

Subjects will be randomised at baseline visit (M0) using a centralised online procedure into one of 2 parallel arms with a 1:1 ratio between the “vaccine” arm and the “Control” arm. Randomisation will be performed by the site staff using the centralized tool in the e-CRF (https://cleanweb.aphp.fr/Ctms-02/portal/login).

Randomisation will be stratified according to the child's VZV serology status (+ or -) at the leukaemia diagnosis . If serology is positive and the child has received a transfusion or intravenous immunoglobulin (IVIg) infusion within the preceding month, a documented history of varicella must be confirmed before considering the child as VZV-positive. If serology is negative, it must be verified that the child has not developed a VZV infection since the initiation of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization. The lists will be prepared in advance by the team of statisticians in the CRU, balanced by permutation blocks of variable size and not communicated to the investigators.

Commenté [CoD65]: At diagnosis supprimé ici mais présent 1ere ligne page 25

Commenté [SE66R65]: Commentaire ajouté page 25 : A confirmer

Commenté [CoD67]: At diagnosis supprimé ici mais présent 1ere ligne page 25

Commenté [SE68R67]: Commentaire ajouté page 25 : A confirmer

Commenté [CP69]: A harmoniser

5. IMPLEMENTATION OF THE STUDY

Before any examination or intervention related to the study may be carried out, the investigator must obtain the *freely given, informed and written consent* of the holder(s) of legal authority and the participant if possible.

Individuals liable to participate in studies benefit from a preliminary medical examination adapted to the study.

The start of the clinical trial is the inclusion of the first patient.

All visits will be performed by physicians involved in the management of patients and trained in Good Clinical Practice for clinical trials, during a standard of care follow-up visit.

Study days should be calculated based on the date of the first vaccination (M0)/ randomisation. One month will be equal to 28 days in this protocol. Visits should be performed ± 8 days compared to reference visit (first vaccination/randomisation) for M1, M2 M6 and M12 visits.

At M0 and M1 visits, participants will receive diary card which are the source document for the follow-up monitoring. The diary card will be collected by the investigator one month after M1 and M2 visits. AE will be graded by the investigator according to protocol gradation scale and will be entered in the eCRF. If after review of events by the investigator, there are changes in the information recorded by the parents, with parental authority, or the patient (if possible) on the diary card, then the investigator must note the correct information in the medical file and justify the correction. The completed diary card will be kept at the site. “Vaccine” Group’s participants will be provided with a ruler for daily measurement of the size of local reactions at vaccination sites, and with a thermometer for daily measurement of temperature. In the event of occurrence of a fever, a medical examination must be carried out as a matter of urgency, as provided for in standard care.

5.1. Screening visit

The screening visit takes place within 3 months preceding the baseline visit.

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During this visit, investigators will propose to the holder(s) of legal authority and the patient that the latter take part to the study and will explain them the study. A copy of the information form will be given for reading, taking time for reflexion and for having further information about the study.

During this visit, the physician will prescribe a VZV serology test (which must be performed during a blood test already scheduled as part of the child's follow-up for leukaemia management) within the month preceding the baseline visit.

The procedure for informing participants and obtaining their consent is described in detail in the section: Ethical and legal considerations.

Whose consent must be obtained	**Who informs the individuals and collects their consent**	**At what point the individuals are informed**	**At what point the consent is obtained**
• The holder(s) of legal authority; • The participant's agreement to take part in research involving the human person is sought as far as possible. (For pre-adolescents and teenagers, no signature is required, but they can be offered to sign this document if they wish)	• the principal investigator or collaborating physician declared and trained in the study (Haematology and Paediatric Oncology)	• Screening visit (standard of care visit)	• After a reflection period of at least 72 hours at the time a specific procedure for the study is implemented (M0)

5.2. Baseline visit M0: Inclusion ~~and randomisation visit~~ ~~(for phase 2)~~

The inclusion visit will represent Month 0 (M0). This visit takes place within a standard care visit.

After parents, holder(s) of legal authority, have consented and patient agrees (if possible) to participate to the study, the informed consent form (ICF) will be signed by holder(s) of legal authority and patient (if he agrees) and the investigator. The signed ICF must be obtained prior to initiating any study specific assessments.

During this visit, the following procedures will be performed:

- Demographic data
- Medical history
  - ALL subtype
  - Date of ALL diagnosis

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- ALL treatment protocol
- Date of the start of induction treatment
- Date of complete remission
- Date of the start of maintenance treatment
- History of clinical varicella and date
- Use of polyvalent or specific immunoglobulins in the past 3 months and date
- Any medical history that may be relevant to participant eligibility for his participation to the study such as prior vaccinations, concomitant medications, and previous and ongoing illnesses (ex: known VHB/ VHC infection) or injuries or other chronic disease
- Any other significant medical conditions which may impair the assessment of the study.

➢ Review within one-month contraceptive methods in women of childbearing age.

➢ Results from standard of care assessment: VZV serology within the previous month; and at M0: CBC, ionogram (Na, K, Ca, Mg, Ph), urea, creatinine, hepatic assessment (AST and ALT, gamma GT, Total serum bilirubin) IgG, IgA and IgM, T, B, NK lymphocyte typing.

➢ Clinical examination with measurement of vital signs (temperature, blood pressures, Sa02 and pulse), weight and height.

During this visit, the medical acts specific to the research are as follows:

➢ Check all contraindications regarding the vaccination

➢ Check if there was a VZV exposure/varicella in the 21 days

➢ Check the VZV status at diagnosis of leukaemia (+ or -). If serology is positive and the child has received a transfusion or intravenous immunoglobulin (IVIg) infusion within the preceding month, a documented history of varicella must be confirmed before considering the child as VZV-positive. If serology is negative, it must be verified that the child has not developed a VZV infection since the initiation of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization.

➢ A urine pregnancy test for women of childbearing age. The physician will ensure that beta-HCG test is negative.

➢ Blood sample for research immunological assessment (ELISA, ELISPOT)

➢ Provide the participant with a *patient card* to be kept with them at all times

➢ **Phase 2 only:** Randomisation to one of the 2 following groups:
  - Group1: “Vaccine Group”
  - Group2: "Control Group”

➢ Give to the participant and their holder(s) of legal authority

➢ Give to the participant and their holder(s) of legal authority

Commenté [SE70]: Question de Christine : Lequel ?

Commenté [CoD71R70]: Si c’est la seule donnée biologique du jour de la visite qui est nécessaire pour la randomisation ou la vaccination alors plutôt un test urinaire pour eviter de faire attendre le patient . a voir avec le deroulé habituel de la visite je pense.

Commenté [SE72R70]: Nous pouvons choisir test de grossesse urinaire uniquement.

Commenté [LGC73R70]: Oui tout à fait possible pour un test urinaire uniquement

Commenté [CP74]: A harmoniser

a mis en forme : Police :Italique

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- Explain to the patient and their holder(s) of legal authority, how and how often the diary card should be filled (DiCa)
    - Phase 1 patients and "Vaccine Group": in order to collect safety information during 7 days after vaccine administration of local (pain, swelling, redness, pruritus at the injection site) and general solicited reactions (chills, fever, fatigue, malaise, myalgia, headache, abdominal pain, diarrhoea, nausea and vomiting) and also concomitant treatments and all other events occurring between study visits.
    - "Control Group": to collect concomitant treatments and all other events occurring between study visits.
- This diary card will be collected at M1 visit.

➢ At the end of the visit, the next visit will be scheduled.

Phase 1 patient and "Vaccine group"

➢ Intramuscular injection of Shingrix®

➢ Monitoring during 30 minutes after vaccination. At the end of this period, the investigator will inspect the site of vaccine administration for cutaneous reactions and ask specific questions regarding local (pain, swelling, redness, pruritus at the injection site) and general symptoms (chills, fever, fatigue, malaise, myalgia, headache, abdominal pain, diarrhoea, nausea and vomiting), and measure vital signs (temperature, blood pressure, Sa02, pulse).

➢ At the end of the visit, the investigator or authorized study staff will give:

- a ruler to measure any skin reactions
- a digital thermometer.

The next visit, teleconsultation, will be scheduled.

5.3. M0+7days teleconsultation (Phase 1 patient and "Vaccine group" only)

Teleconsultation will be carried out 7 days following the first vaccine administration to record, with the help of the diary card, any adverse events that have occurred since the injection.

5.4. Visit M1

This visit will take place at Month 1 (Day 28) ± 8 days, within a standard care visit.

During this visit, investigator or authorized study staff will:

➢ Review of concomitant medication: new treatments and changes in treatment since the previous visit (with diary card as support).

➢ Record of AEs and SAEs: local and systemic vaccine related events (any grade) ("Vaccine" group); any grade other event (with diary card as support).

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- Check the results from standard of care assessment: CBC ionogram (Na, K, Ca, Mg, Ph), urea, creatinine, hepatic assessment (AST and ALT, gamma GT, Total serum bilirubin), IgG, IgA and IgM, T, B, NK lymphocyte typing
- Check for possible VZV infection or VZV exposure since the first injection.
- Clinical examination with measurement of vital signs (temperature, blood pressures, pulse, Sa02, FR), weight and height
- Collect from the participant and their holder(s) of legal authority the first diary card (DiCa).
- Give to the participant and their holder(s) of legal authority a new diary card (DiCa) that should be filled until the next visit. This diary card will be collected at M2 visit.
- Blood sampling (before vaccination where applicable) for research immunological assessment (ELISA)
- Schedule the next visit

Phase 1 patient and "Vaccine Group":

- Check all contraindications regarding the vaccination
- Perform injection of Shingrix®
- Perform monitoring for 30 minutes after vaccination: At the end of this period, the investigator will inspect the site of vaccine administration for cutaneous reactions and ask specific questions regarding local (pain, swelling, redness, pruritus at the injection site) and general symptoms (chills, fever, fatigue, malaise, myalgia, headache, abdominal pain, diarrhoea, nausea and vomiting), and measure vital signs (temperature, blood pressure, pulse, Sa02, FR).

5.5. M1+7 days teleconsultation (Phase 1 patient and "Vaccine group" only)

Teleconsultation will be carried out 7 days following the second vaccine administration to record, with the help of the diary card, any adverse events that have occurred since the injection.

5.6. M2, M6 Follow-up visits

5.6. M2, M6 Follow-up visits

These visits will take place respectively at Month 2 (Day 56 ± 8 days), at Month 6 (Day 168 ± 8 days).

During this visit, investigator or authorized study staff will:

- Collect information, from standard of care, about the medical history of the patient
  - Record of new treatments taken and changes in treatment since the previous visit (with diary card as support).

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- Record any significant medical events: local and systemic vaccine related events (any grade) (“Vaccine” group); any grade other event (with diary card as support).
- Results from standard of care assessment at M2, M6: CBC, ionogram (Na, K, Ca, Mg, Ph), urea, creatinine, hepatic assessment (AST and ALT, gamma GT, Total serum bilirubin), IgG, IgA and IgM, T, B, NK lymphocyte typing.
- Check for possible VZV infection or VZV exposure since the previous visit
- Clinical examination with measurement of vital signs (temperature, blood pressures, pulse, Sa02, FR), weight and height

➢ Blood sample for research immunological assessment (ELISA and ELISPOT [only at M2 for ELISPOT])

➢ Collect from the participant and their holder(s) of legal authority the second diary card (DiCa) (only M2 visit)

➢ Schedule the next visit

5.7. Last study visit

The visit at M12 (Day 336 ± 8 days) will be the last visit of the research.

During this visit, the standard of care examinations will be performed:

➢ Clinical examination with measurement of vital signs (temperature, blood pressures, pulse, Sa02, FR), weight and height

➢ Review of concomitant medications taken and changes in treatment since the previous visit, related medical reason.

➢ Record any significant medical events (any grade).

➢ CBC, ionogram (Na, K, Ca, Mg, Ph), urea, creatinine, hepatic assessment (AST and ALT, gamma GT, Total serum bilirubin), IgG, IgA and IgM, T, B, NK lymphocyte typing.

➢ Check for possible VZV infection or VZV exposure since the previous visit

Blood sampling will be performed for research immunological assessment (ELISA, ELISPOT).

At the end of the study, participants will continue their standard of care.

5.8. Data collection in case of VZV infection or VZV reactivation

In the event of a VZV infection/reactivation, the following data will be collected from standard of care:

- PCR results (if available)
- The nature and source of the infection
- Severity (hospitalisation, admission to intensive care),
- Duration of treatment,
- Time to resumption of chemotherapy

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5.9. Early termination visit

In case of early termination of trial follow-up (e.g., withdrawal from trial follow-up prior to M12), every effort should be made to perform the procedures of the end of research visit (equivalent to M12 visit) at the time of withdrawal.

Reasons for early termination of follow-up should be documented.

Participant will then continue medical care as scheduled in the follow-up of the pathology.

5.10. Expected length of participation and description of the chronology and duration of the study.

The duration of participation for each participant will be 12 months, whereas the duration of recruitment will be 30 months. Overall, the total duration of the study will be 42 months (FVFP to LVLP).

Phase 2: Participants will be randomised during the baseline visit (M0).

Participants will receive best available routine care for the duration of the study.

Duration of enrolment period	30 months
The length of participation for participants, of which:	12 months
• Treatment duration:	1 month (Phase 1 patient and "Vaccine" group only)
• Duration of follow-up period:	11 months for Phase 1 and "Vaccine group", and 12 months for "Control" group
Total study duration:	42 months

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5.11. Table or diagram summarising the chronology of the study

Actions	**Screening visit** (M-3 to D-1)	**M0 (Baseline visit)**	**M0+7 days** ± 21 day	**M1** ± 8 days	**M1+7 days** ± 21 day	**M2** ± 8 days	**M6** ± 8 days	**M12** ± 8 days (End of
Information	X							
Informed consent		X						
Verification of inclusion and exclusion criteria		X						
Medical History	X	X						
Standard of care clinical examination	X	X		X		X	X	X
Paraclinical examinations (temperature, Sa02, FR, blood pressure, pulse)	X	X		X		X	X	X
Pregnancy test (urinary blood)		X						
Collection of VZV status at diagnosis of leukemia*within the previous month		X						
Collection date from standard of care: CBC platelets, ionogramm (Na, K, Ca, Mg, Ph), urea, creatinine, hepatic assessment (AST and ALT, gamma GT, Total serum bilirubin) IgG, IgA and IgM, T, B, NK lymphocyte typing, VZV serology.		X		X		X	X	X
Randomisation		X						
Check of vaccination contraindications (phase 1 patient and "Vaccine" group)		X		X				
Shingrix® administration (phase 1 patient and "Vaccine" group)		X		X				
Post-vaccination monitoring (phase 1 patient and "Vaccine" group)		X		X				
Diary card (DiCa)** provided		X (N°1)		X (N°2)				
Diary card (DiCa)** reviewed and collected				X (N°1)		X (N°2)		
Teleconsultation (phase 1 patient and "Vaccine" group)			X		X			
Adverse events **		X	X	X	X	X	X	X
Blood samples (ELISA, ELISPOT)		X				X		X
Blood Samples (ELISA)				X			X	
Anti-gE and anti-VZV response evaluation		X		X		X	X	X
Cellular immune response evaluation (gE and VZV)		X				X		X

*: If serology is positive and the child has received a transfusion or intravenous immunoglobulin (IVIg) infusion within the preceding month, a documented history of varicella must be confirmed before

Commenté [CoD75]: Je pense qu'il faut être un peu souple pour eviter des déviations

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considering the child as VZV-positive. If serology is negative, it must be verified that the child has not developed a VZV infection since the initiation of leukemia treatment; if such an infection occurred, serology will be reassessed at the time of randomization.
**: Collection of AEs includes for the Phase 1 patient and “Vaccine group”, solicited reactions for 7 days (M 0 to M0+7 days, M1 to M1+7 days) and unsolicited adverse reactions for 30 days (following each vaccination) and for both groups any grade adverse event (with the help of diary cards for the first two months). N°1 and N°2 correspond to Diary cards to complete.

5.12. Distinction between standard care and study

TABLE: "Standard care" vs. "additional interventions" required specifically for the study

Interventions, procedures and treatments carried out for research purposes	**Interventions, procedures and treatments associated with standard care**	**Interventions, procedures and treatments added for research purposes**
Treatments		Shingrix® administration at M0 and M1 (“Vaccine” group only) and Post-vaccination monitoring
Visits	All medical visits	Teleconsultation at M0+7 days and M1+7days (“Vaccine” group only)
Blood samples		Additional tubes for research analyses for both groups: ELISA at M0, M1, M2, M6 and M12; ELISPOT at M0, M2 and M12
Clinical and biological data	The data are collected from medical file	
Reactogenicity to vaccine (Phase 1 patient and “Vaccine group”) and AE collect		Diary card (DiCa)

Commenté [LGC76]: Est-ce des tubes complémentaires ou des prélèvements complémentaires ?
Merci de clarifier svp

5.13. Biological samples collection

- Serum preparation: 2 x 3.5 ml of blood will be taken. Serum will be aliquoted into 2 cryotubes of at least 1mL each (for Anti-gE and for anti-VZV response evaluation).
    - For centres inside Paris, blood tubes will be sent directly to the laboratory of immunology at Saint Antoine that will be in charge of centrifugation, serum extraction and storage at -20°C.
    - For centres outside Paris, serum samples will be prepared and frozen by the CBRs until transportation to dedicated laboratories.
- PBMC for ELISPOT: tubes of blood on lithium heparinate will be taken to prepare a minimum of 1 cryotube of 5 million of PBMC, according to the procedure. The volume of blood to be drawn will depend on the child's weight (for examples: 6-15kg: 3-5mL; 15-30kg: 5-8.5mL, >30kg: 8.5 to 15mL).

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- For centres inside Paris, blood will be sent directly to the Necker laboratory that will perform preparation and congelation of PBMC.
- For centres outside Paris, PBMC will be prepared by local CBR then stored at -150°C or in liquid nitrogen by the CBR until transportation to dedicated laboratory.

The sample will be pseudonymised and sample handling will be documented.

Transportation of serum and PBMC samples to dedicated laboratories will be conducted by a carrier chosen by the management centre, according the following calendar:

- Firstly, as soon as patients have performed the M2 visit and secondly at the end of the study for serum samples
- At the end of the study (last M12 visit) for PBMC samples
- At sponsor request

Serum will be analysed by Virology Laboratory– Federative Institute of Biology, Toulouse (responsible: CHAPUY REGAUD Sabine) to evaluate anti-gE IgG and by the department of virology, Hôpital Saint-Antoine GHUEP (responsible: Pr. Morand-Joubert Laurence) to measure neutralizing anti-VZV IgG.

ELISpot will be performed at the laboratory of immunology, Necker hospital, Paris (Responsible: Dr Le Gall-Roelens Marie).

Samples will be then stored in theses laboratories until the end of the trial.

At the end of the study, the samples will be destroyed.

Type of sample	**Quantity**	**Storage location** **(Name and entity)**	**Supervisor of the sample collection (Name and entity)**	**Purpose of the sample collection**	**Storage duration**	**End use/Future (e.g., destruction) after the end of the CT**
Serum	About 1 mL	Virology Laboratory– Federative Institute of Biology Toulouse	Virology Laboratory– Federative Institute of Biology: Dr. CHAPUY REGAUD Sabine	anti-gE IgG ELISA	Until the end of the CT	destruction after the end of the CT
Serum	About 1 mL	Department of virology, Hôpital Saint-Antoine GHUEP Paris	Department of virology, Hôpital Saint-Antoine GHUEP: Pr. MORAND-JOUBERT Laurence	anti-VZV IgG ELISA	Until the end of the CT	destruction after the end of the CT

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Type of sample	Quantity	Storage location (Name and entity)	Supervisor of the sample collection (Name and entity)	Purpose of the sample collection	Storage duration	End use/Future (e.g., destruction) after the end of the CT
PBMC	About $5x10^6$ PBMCs	Laboratory of immunology at Necker Paris	Laboratory of immunology at Necker: LE GALL Marie	ELISpot	Until the end of the CT	destruction after the end of the CT

6. ELIGIBILITY CRITERIA

6.1. *Inclusion criteria*

1. Children aged 1 to 17 years with acute lymphoblastic leukaemia
2. Inclusion within 6 months of the start of maintenance treatment
3. Participants with childbearing potential* having reliable contraception from 4 weeks before vaccination to 12 weeks after the last vaccination, such as established use of oral, injected or implanted hormonal methods of contraception; placement of an intrauterine device (IUD) or intrauterine system (IUS); barrier methods: condom or occlusive cap (diaphragm or cervical/vault caps) with spermicidal foam/gel/film/cream/suppository; male partner sterilization (the vasectomized partner should be the sole partner for that subject); surgical sterilization (hysterectomy, bilateral oophorectomy, tubal ligation) or true abstinence (when this is in line with the preferred and usual lifestyle of the subject) prior to enrolment at M0
4. Patient affiliated to a social security scheme
5. Informed consent of holder(s) of legal authority and of the participant if applicable

*A woman is considered of childbearing potential (WOCBP), i.e. fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilisation methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. A high follicle stimulating hormone (FSH) level in the postmenopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient.

6.2. *Exclusion criteria*

1. History of varicella exposure/varicella in the 21 days prior to randomisation
2. Platelet count < 50 G/L or PNN count < 0.5G/L at the time of randomisation
3. Any bleeding disorder considered a contraindication to intramuscular injection
4. Treatment with current or prior exposure to Inotuzumab
5. Vaccination against influenza within 15 days and for other vaccines within 4 weeks prior to the randomisation up to M2
6. Fever within 72 hours before randomisation

a mis en forme : Police :11 pt

Commenté [LGC77]: Ajout de la définition d'une femme en âge de procréer demandée systématiquement par l'ANSM à présent

a mis en forme : Police :11 pt

a mis en forme : Police :11 pt

a mis en forme : Police :11 pt

a mis en forme : Police :11 pt

a mis en forme : Retrait : Gauche : 0 cm, Première ligne : 0 cm

Commenté [LGC78]: Ajouter femmes enceintes ou allaitantes ? Purement réglementaire

Commenté [CP79R78]: ok

Commenté [FAL80R78]: je l'aurai pas mis ca n'a pas de sens dans ce contexte d'enfant en cours de traitement d'une LAL

Commenté [CPE81]: Juste une question pour ma compréhension personnelle : Est-ce par précaution ? ou lié à l'immunosuppression générée par le traitement lui-même pouvant diminuer la réponse au vaccin ? car Inotuzumab n'est pas recommandé avec vaccins viraux vivants mais il n'y a rien dans le RCP par rapport aux vaccins recombinants comme shingrix

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7. Allergy to one of the components of the vaccine or serious allergic history (anaphylactic type)
8. History of severe adverse reactions following administration of a vaccine, including an anaphylactic reaction and associated symptoms such as rash, difficulty breathing, angioedema and abdominal pain, or a history of allergic reaction that may be exacerbated by a component of the vaccine.
9. Known infection with HIV, HCV or HBV
10. Current or planned participation in a vaccination trial
11. Participation without the investigator's agreement in other research within 4 weeks prior to the randomisation and for the duration of the trial.

6.3. Recruitment procedure

Recruitment will take place in departments of Paediatric, Haematology, Immunology, and Oncology of 6 sites. During hospital appointments, investigators will recruit patients from among those they are treating for their condition.

	Number of participants
Total number of participants to be included	78 Phase 1: 12 Phase 2: 66
Number of centres	6 Phase 1: 1 centre* Phase 2: 6 centres
Enrolment period (months)	30
Number of participants/centre	10 to 30
Number of participants/centre/month	**0.7 to 2.1**

** For Phase I, the Trousseau centre will take part; if recruitment 'is slow', we may involve other APHP centres. All centres that have confirmed their participation will take part in Phase II.*

Commenté [PC82]: Précision nous au moins 1 centre afin d'éviter une MS ?

6.4. Termination rules

6.4.1. Termination rules for individual participants

6.4.1.1. Criteria and procedures for temporary suspension of the study treatment and continuation of the participation from the study

Temporary suspension of the treatment is allowed, especially to recover from adverse events. The treatment can be resumed once the adverse event has been resolved.
If, during the course of his/her participation in the study, the participant presents one of the following adverse events (including exclusion criteria), the study product/procedure must be temporary discontinued but the participant will continue to be monitored for the study.

The administration of VZV vaccine should be postponed in the following situations

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Condition	Rescheduling of vaccination	Follow-up
Minor infection, such as a cold	No deferral	Follow-up will continue as schedule in the protocol.
Acute severe febrile illness, body temperature >38.0°C within 72 hours prior vaccination	Vaccination visit could be rescheduled, within the authorised interval (8 days).	Participant will continue to be monitored for the study and all procedures are applied.
Insufficient haematological conditions (post-chemotherapy): PNN<0.5G/L and/or platelets< 50G/L	Administration of the vaccine is postponed until PNN≥0.5G/L and platelets ≥50 G/l are obtained.	The next visit is rescheduled and carried out no earlier than 4 weeks after the injection. All procedures are applied. Subsequent visits will be carried out according to the initial schedule.

The investigator must document the reason for delaying the treatment in the participant's source file.

In the case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the severe adverse events are resolved. Notification of a serious adverse event must be sent by email (eig-vigilance.drc@aphp.fr) to the sponsor. If a Data Safety Monitoring Board has been created, the committee can specify and/or validate the follow-up methods.

6.4.1.2. Criteria and procedure for definitive withdrawal of the study treatment and continuation of the participation from the study

Condition	Vaccination	Follow-up
Occurence of a grade ≥ 3 adverse event related to vaccine (TDL) or in the occurrence of a clinical allergy to one of the components of the vaccine after the 1st injection	No vaccination (and no teleconsultation)	The subject will remain in the study (with protocol samples) until the end of his/her participation except in the situations described below (see below section "Criteria and procedure for exit the study").
Occurence of chickenpox infection or reactivation of shingles	No vaccination (and no teleconsultation)	The patient will remain in the study. The next visit (M2) is scheduled 1 month after infection, then at M6 and M12 (with protocol samples).

The investigator must:

Commenté [CoD83]: Est-ce qu'il faut clairement cités les TDL pour la phase 1 ?

Commenté [FAL84R83]: peut etre voir dans le protocole PACIFIC ce qu'on avait mis

Commenté [CP85]: à confirmer avec S.Chevret pour redescendre de dose ?

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- Document the reason(s) for discontinuation the study treatment in the subject's source file and CRF.
- In case of SAE, notify the sponsor and follow up the participant until the serious adverse events are resolved.
- Collect the evaluation criteria at the time of stopping vaccination in the research

6.4.1.3. Criteria and procedure for exit the study

- **Criteria**

- Medical Issue: The investigator can permanently withdraw a participant from the study for any safety reason (ex: pregnancy or serious adverse event) or if it is in the participant's best interests.
- Personal reasons of the participant: Participants may exit the study at any time and for any reason.
- Explicit withdrawal of consent
- Lost to follow-up
- Participant's death

- **Procedures**

If a participant exits the study, the investigator must notify the reason in medical file and complete in the case report form the various reasons why the participant has discontinued the study. Premature exit from the study of a participant will not affect the participant's ongoing care.

- Medical Issue

In case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the severe adverse events are resolved.

- Personal reasons of the participant

If a participant exits the study prematurely, not to withdraw consent, any data collected prior to the date of premature exit may still be used.

- Explicit withdrawal of consent

If a participant exits the study prematurely, withdraws consent and objects to the use of his/her data, any data collected prior to the date of premature exit may still be used.

- Lost to follow-up

The participant cannot be located. The investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead (for example: to get in contact with the birth city hall).

Commenté [CoD86]: Ne faut il pas ajouter participant's death (car n'est pas evoqué ici or il fait partie des critères ci-dessus)

6.4.2. Termination rules for the clinical trial: full or partial discontinuation of the clinical trial

AP-HP as sponsor or the Competent Authority can prematurely discontinue all or part of the study, temporarily or permanently, further to the recommendations of the Data Safety Monitoring Board if suspected unexpected serious adverse reactions (SUSARs) are observed in the treatment arms or if there is a discrepancy in the serious adverse reactions between the arms, requiring a reassessment of the benefit-risk ratio for the study.

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Similarly, AP-HP, as the sponsor, or the Competent Authority may decide to prematurely discontinue the study due to unforeseen issues or new information about the product, in light of which the objectives of the study or clinical programme are unlikely to be achieved**.**

AP-HP as sponsor reserves the right to permanently suspend enrolment at any time if it appears that the inclusion objectives are not met.

If the study is prematurely discontinued, the decision and justification will be provided by the sponsor (AP-HP) to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but not later than in 15 days of the date of the temporary halt or early termination.

If the study is prematurely discontinued for safety reasons, it shall include the reasons for such action and specify follow-up measures, along with recommendations from the Data Safety Monitoring Board in the case of substantial modification.

In case of a protocol substantial modification is necessary, it should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but not later than 15 days of the date of the temporary halt or early termination.

In case of occurrence of an unexpected event that could have a serious impact on the risk/benefit of the trial, this unexpected event should be shared without undue delay but not later than 15 days from the date the sponsor became aware of this event.

In case of implementation of urgent safety measures, they should be taken immediately and should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but no later than 7 days of the date of the measures implementation.

7. TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

Commenté [SE87]: @ AGEPS

7.1. Description of the investigational medicinal product(s)

The investigational medicinal product of the study is HZ/su vaccine (Shingrix®) (GSK). The reference document for the test product will be the Summary of Product Characteristics (SmPC).

Presentation:

One commercial pack contains:

- 1 vial of powder with brown removable cap (antigen)
- 1 vial of suspension with a removable blue-green cap (adjuvant), for one dose = 0.5 mL

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Shingrix® is indicated for the prevention of shingles and postherpetic neuralgia (PHN) in:

- adults aged 50 or over;
- adults aged 18 or over at increased risk of shingles.
- ~~In this study, vaccine will not be used in their market authorization indication in order to adapt it to children. Therefore, powder will be suspended with only 0.25 mL of the suspension of the adjuvant.~~

Preparation of Shingrix®:

It will be reconstituted prior to administration in accordance with recommendations detailed in the SmPC. The vial should be gently shaken until the powder is completely dissolved.

- If the reconstitution is made by the pharmacy of each clinical site (preferably), the shelf life of the solution from reconstitution to injection will be the one reported by the pharmacy on the labelling of the syringe.
- If the reconstitution is made by the clinical department, the vaccine will be prepared extemporaneously and it must be administered immediately after reconstitution.

Administration should be done by intramuscular route only.

After reconstitution, one dose (0.5 mL) contains:

- Varicella-Zoster Virus glycoprotein E antigen: 50 micrograms
- Adjuvant AS01B containing:
  - Quillaja saponaria Molina plant extract, fraction 21 (QS-21): 25 micrograms
  - 3-O-desacyl-4'-monophosphoryl lipid A (MPL) from Salmonella minnesota: 25 micrograms

Administration schedule:

During the study, each patient will receive a regimen consisting of 2 intramuscular injections of Shingrix® HZ/su vaccine 1 month apart (M0 and M1) at the dose corresponding to the trial phase, as described below and the first dose will be given within 6 months of the start of maintenance treatment in children with ALL.

~~Two injections of HZ/su vaccine (Shingrix®) will be given one month apart (M0 and M1) at the dose corresponding to the trial phase, as described below.~~

Doses in the clinical trial:

Phase 1:

Patients will be enrolled sequentially, by cohorts of 2 patients. Once a cohort has been completed, based on the observed endpoint (i.e., occurrence or not of DLT within the first 7 days post each infusion of vaccine) of the 2 patients from this cohort, the estimation procedure is rerun, and recommended dose level for patients in the second cohort will use observed responses (DLT or not) of each previously enrolled patient. For ethical concerns, no skipping will be allowed so that no dose level could be recommended unless all previous levels have been tested. The process is continued until the sample size of 12 patients has been reached.

a mis en forme : Retrait : Gauche : 0,63 cm, Suspendu : 0,63 cm, Espace Avant : 2 pt, Avec puces + Niveau : 1 + Alignement : 0,63 cm + Retrait : 1,27 cm

Commenté [CoD88]: Je mettrai laphrase après ou pas du tout car ici on dit qu'on ne suit pas l'autorisation et le paragraphe suivnat on decrit une reconstitution « normale »

Commenté [SE89R88]: Effectivement, à revoirEffectivement, à revoir
@AGEPS

a mis en forme : Police :Non Gras, Non Italique, Soulignement

a mis en forme : Police :Non Gras, Non Italique, Soulignement

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HZ/su vaccine (Shingrix®) will be administered by injection at the following dose levels (DL):
DL 1: 1/2 dose of the reconstituted Shingrix® vaccine, i.e 0.25 mL
DL 2: 3/4 dose of the reconstituted Shingrix® vaccine, i.e 0.38 mL
DL 3: full dose of the reconstituted Shingrix® vaccine, i.e 0.5 mL

Phase 2

HZ/su vaccine (Shingrix®) will be administered by injection with respect to the recommended dose level agreed following the phase 1.

~~Shingrix® must be reconstituted before administration. The vial should be gently shaken until the powder is completely dissolved.~~

~~If the reconstitution is made by the pharmacy of each clinical site (preferably), the total shelf life of the solution from reconstitution to injection should not exceed 6 hours at +2 +8°C.~~

~~If the reconstitution is made by the clinical department, the vaccine will be prepared extemporaneously and it must be administered immediately after reconstitution.~~

~~Administration should be by intramuscular route only.~~

Supply:

Shingrix® will be provided by clinical trial department of AGEPS (Agence Générale des Equipements et Produits de Santé) that will ensure the secondary labelling according to the good manufacturing practices, and the supply to the pharmacy of each clinical site according to the good distributing practices.

Special precaution for storage:

The product should be stored at +2 to +8°C in the original package in order to protect it from light. It should not be freezed.

Commenté [CA90]: Etes-vous d'accord pour arrondir à 0.38 mL ? le volume pour ¾ de dose serait de 0.375 mL mais les seringues de précision ne permettent qu'une adaptation du volume à 0.01 ml près

Commenté [CP91R90]: Volume à valider par Fanny

Commenté [FAL92R90]: je pense qu'il faut mettre 0.3, 0.4 et 0.5 mL pour que ce soit plus simple

a mis en forme : Normal, Espace Avant : 6 pt, Sans numérotation ni puces

a mis en forme : Normal, Espace Avant : 6 pt, Sans numérotation ni puces

a mis en forme : Espace Après : 0 pt

a mis en forme : Espace Après : 0 pt

7.2. Description of Auxiliary medicinal product (treatment required to conduct the study

According to the Regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014, an auxiliary medicinal product is a medicinal product used in the context of a clinical trial but not as investigational medicinal product, as described in the protocol for background treatments, as challenging agents, rescue medication or to assess the end-points.

In this clinical trial, the auxiliary medicinal product used according to their market authorization is: EMLA PATCH 5%.

EMLA PATCH is a 10 cm² adhesive skin plaster containing 1 g of emulsion composed of lidocaine (25 mg) and prilocaine (25 mg). EMLA is a topical anaesthetic for healthy skin during needle insertion or superficial surgical procedures that can be used in adults and children.

EMLAPATCH should be applied at least 1 hour before vaccination.

The reference safety information applicable to the auxiliary medicinal product is the SmPC.

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7.3. Description of traceability elements accompanying the investigational medicinal product(s)

A patient card will be given to all patients, bearing the words "Please keep this card with you at all times" and specifying the name, address and telephone number of the investigating centre, the research code, the patient identification code and the dates ~~began.~~of administrations.

A pharmacy manual will describe supplies, storage, dispensing, administration, accountability and destruction will be provided to the pharmacy of each centre during the opening visit.

The pharmacist of the healthcare establishment is responsible for ensuring adequate storage conditions and proper management of the investigational products, including confirming that each shipment has been received correctly both quantitatively and qualitatively. The local pharmacy will be in charge of accounting, traceability of investigational drugs, which will be checked regularly by the CRA monitoring the study.

An accounting system for medical product will be set up at each research centre in order to trace:
- On receipt: date, quantities, batch numbers and the expiry dates.
- On dispensing, for each patient, according to the patient identification code: dispensing date, quantities dispensed, batch numbers and expiry dates.

Dispensing will be made on presentation of a specific nominative prescription generated by the electronic CRF, in accordance with local procedures.

The sponsor will provide pharmacy departments with traceability document for each vaccine dispensing: name, batch number, number of doses delivered. Pharmacy or clinical department in charge of the reconstitution will record this step on own traceability documents. Clinical staff will report all information about the vaccination (name, batch number, administration route, time of administration) in the medical file.

7.4. Authorised and prohibited treatments (medicinal, additional medicinal, surgical), including rescue medications

It is not authorised in the context of this research to administer any other vaccine within 28 days (except for the influenza vaccine within 14 days) before or after the trial vaccine doses.

In the event of anticoagulant treatment with low molecular weight heparin, its injection should not be carried out either the day before or the day of the intramuscular injection of the vaccine. It may be resumed at least 4 hours after the vaccine has been administered. The benefit-risk balance of this change in treatment will be at the discretion of the investigating physician.

In the event of a varicella exposure, use of polyvalent immunoglobulin or Varitect is authorised in the 10 days following exposure. This treatment must be reported in the CRF.

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7.5. Methods for monitoring compliance with the treatment

Injections of vaccine will be done in consultation or one-day hospitalization by the physicians in charge of the participants.

8. ASSESSMENT OF IMMUNOGENICITY

8.1. Description of immunigenicity endpoints assessment parameters

- Analysis of humoral response:

Humoral response will be evaluated at M0, M1, M2, M6 and M12 visits by two approaches.

- First, humoral immune response will be evaluated by measuring IgG directed to the glycoprotein E of the VZV that is present in the vaccine (primary endpoint).

The Anti-glycoprotein E humoral immune response is defined as a titer of anti-gE IgG antibodies:

- in VZV - subjects at inclusion: > 4 x positivity cutoff (97 mIU/mL)
- in VZV+ subjects at inclusion: > 4 x anti-gE IgG level measured at M0

- Secondly, humoral immune response will be evaluated by measuring IgG directed to the VZV.

The VZV humoral response is defined as a titer of anti VZV IgG antibodies:

- in VZV - subjects at inclusion: > 165 mUI/mL
- in VZV + subjects at inclusion: > 4 x IgG anti-VZV level measured at M0.

- Cellular immune response.

Cellular immune response will be evaluated at M0, M2 and M12 visits by two approaches.

- Assessment of the induction of a CD4+ T vaccine response by *Ex vivo* stimulation with whole gE or with a lysate of VZV-infected cells and measurement by ELISpot (IFNg, IL-2). Induction of CD4+T response will be evaluated at M0, M2 and M12.

As expected for a recombinant subunit vaccine, specific CD4+ polyfunctional T cells, but no specific CD8+ T cells, were reported after Shingrix vaccination (20). Therefore, after vaccination and compared to baseline, PBMC response to ex vivo stimulation by full-length recombinant gE, will reflect immune response in the CD4+ T cell compartment. PBMC response to VZV lysate would allow evaluating a more global cellular immune response against the virus, especially in VZV-seropositive patients (21).

A response was considered positive to a cytokine if the number of spots in wells stimulated with VZV antigens (recombinant gE or VZV lysate) was at least 2-fold greater than the number of spots in the negative control, using a cutoff of 10 SFC/10^5 cells after background subtraction as previously described. Data will be expressed as a number of spots per well.

A positive response to the vaccine was defined as a 2-fold increase in cytokine-producing spike-specific CD4+ T cells at M2 or M12 compared to baseline (M0).

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8.2. Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

- Methods used for the analysis of humoral response:

➢ Humoral response against gE will be performed in the Virology Laboratory - Federative Institute of Biology, Dr. S. CHAPUY REGAUD. Measurement of anti-gE IgG levels will be performed, at all visits (M0 to M12) and realized by an Elisa test (kit distributed by Interchim). An assay range based on the WHO International Standard Anti-Varicella-Zoster Antibody 90/690 or W1044 will be produced.

➢ Humoral response against VZV will be performed in the Department of Virology, Hôpital Saint-Antoine GHUEP, Pr. L. Morand Joubert. Measurement of anti-VZV IgG levels will be performed, at all visits (M0 to M12), by Chemiluminescent Immunoassay using the LIAISON® VZV IgG HT kit on LIAISON® XL (Diasorin). A positive result is defined as: VZV IgG>165 mUI/mL.

- Methods used for assessment of the cellular response:

Cellular response will be performed in the Department of Immunology, Hôpital Necker, Paris, Dr. M. Le Gall-Roelens.

Cellular immune response and its polyfunctionality will be assessed in vitro by measuring IFN-g and/or IL-2 production from VZV-specific T cells using the ELISpot technique (ImmunoSpot, C.T.L., Germany). Peripheral blood mononuclear cells (PBMC) (10^5/well) will be sensitized for 20h at 37°C with the recombinant full-length glycoprotein E (Shingrix immunogenic component), or whole native VZV antigens purified from a cellular lysate (all from Institut Virion/Serion, Germany). Each subject will have a negative control (unstimulated cells in CTL-test medium, Germany) and a positive control (cells stimulated with 10µg/mL PHA, Sigma-Aldrich, France, and/or CEFX Ultra SuperStim Pool, JPT Peptide Technologies, Germany). Post-incubation, plates will be revealed according to the manufacturer's instructions, then scanned and analyzed on a C.T.L reader (S6 Ultimate).

9. SPECIFIC STUDY COMMITTEES

9.1. Steering Committee

- Committee members:

- Dr Fanny ALBY-LAURENT, coordinating investigator (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),
- Chaima MRAD (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),
- Pr. Odile LAUNAY, scientific responsible (Cochin CIC of vaccinology, Cochin hospital, APHP, Paris)
- Pr. Arnaud PETIT, scientific responsible (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris)
- Corinne DESAINT (CIC Cochin of vaccinology, Cochin hospital, APHP, Paris)
- Pr. Sylvie CHEVRET, biostatistician (DRCI, URC Saint Louis, APHP, Paris),
- Representatives of the sponsor (URC and Promotion project managers and DEC-AGEPS).

Commenté [PC93]: A valider par AGEPS

Commenté [CA94R93]: oui

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- Roles:

- Define the overall structure of the study, coordinate information, determine the initial methodology and oversee the study.
- Propose procedures to be followed during the study, acknowledging any recommendations from the Data Safety Monitoring Board, if applicable. The DRCI sponsor retains decision-making authority.

- Operating procedures:

The steering committee will meet according to the state of progress and difficulties encountered in the study.

9.2. Scientific Committee

- Committee members:

This committee consists of:

- Dr Fanny ALBY-LAURENT, coordinating investigator (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),
- Chaima MRAD (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),
- Pr. Odile LAUNAY, scientific responsible (CIC Cochin of vaccinology, Cochin hospital, APHP, Paris),
- Pr. Arnaud PETIT, scientific responsible (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),
- Corinne DESAINT (CIC Cochin of vaccinology, Cochin hospital, APHP, Paris),
- Pr. Sylvie CHEVRET, biostatistician (DRCI, URC Saint Louis, APHP, Paris),
- Emmanuelle SACCO (DRCI, URC Saint Louis, APHP, Paris),
- Dr. Marion STRULLU (Pediatric Hematology-Immmunology Department, Robert Debré Hospital, APHP, Paris),
- Pr. Sabine CHAPUY-REGAUD (Virology Laboratory - Federative Institute of Biology, Toulouse)
- Dr. Claire Dubois (Department of Virology, Hôpital Saint-Antoine GHUEP, Paris),
- Dr. Marie LE GALL-ROELENS (Department of Immunology, Hôpital Européen Georges-Pompidou, Paris).

- Roles

- Define the purpose,
- Draft the protocol,
- Suggest modifications to the protocol during the study.

Roles of this committee are defined in an APHP procedure.

- Operating procedures

This committee will meet at least once a year. A report will be written after each meeting and communicated to all members.

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9.3. Endpoint Adjudication Committee

- Committee members:

Dr Fanny Alby-Laurent and Pr. Arnaud Petit (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris).

- Roles:

To validate the outcomes which are important for the secondary endpoint definitions (causal relationship, grade of events): The committee will make the review of the grade ≥3 adverse events or reactions and the failures of vaccination.

- Operating procedures:

A specific procedure will be written and presented to members at the beginning of the study.

9.4. DLT Evaluation Committee

- Committee members:

– Pr. Arnaud PETIT, scientific responsible (Pediatric Hematology, Immmunology and Oncology Department, CHU Armand Trousseau, APHP, Paris),

–

- Roles:

This committee will be consulted as soon as the DLT is declared. It must respond within 24 working hoursand submit the information , to the Biostatistics and Medical Information Department (headed by Prof. Sylvie Chevret), which is responsible for running the algorithm (which will have been programmed in advance) to determine the dose level that the next patient enrolled in each stratum should receive.

- Operating procedures:

A specific procedure will be written and presented to members at the beginning of the study.

Commenté [SE95]: –PACIFIC : Dr. Marion STRULLU (Pediatric Hematology-Immmunology Department, Robert Debré Hospital, APHP, Paris),
Qui met-on comme deuxième membre ?

Commenté [CP96R95]: Fanny ?

Commenté [FAL97R95]: Ca peut etre moi ?

Commenté [CoD98]: Il ne regarde pas pas tous les Aes en systématique pour s'assurer qu'il n'ya pas un DLT reporté mais non déclaré ?

Commenté [SE99R98]: L'URC sera en charge de s'assurer que les cas saisis ne sont pas des DLT et feront un retour aux centres si nécessaires.

10. SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY

Commenté [SE100]: @vigilance

10.1. Description of Safety endpoints assessment parameters

- Solicited adverse events from M0 to M0+7days: local (pain, swelling, redness, pruritus at the injection site) and general symptoms (chills, fever, fatigue, malaise, myalgia, headache, abdominal pain, diarrhoea, nausea and vomiting).
- Adverse events related to vaccine injections;
- Others adverse events from M0 to M12 occurring in all participants;
- Serious adverse events from M0 to M12 occurring in all participants
- Number of VZV infections or reactivations occurrence during the trial in all participants.

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10.2. Anticipated methods and timetable for measuring, collecting and analysing the safety endpoints

Safety data will be collected within all visits by interview. Moreover, diary cards will be given after each vaccination to participant and her/his holder(s) of legal authority to complete. The dairy cards will give us an exhaustive record of adverse events occurring during the month after each vaccination. All AE reported in the diary card will be discussed at M1 and M2 visits together with participant and the investigator.

10.3. Recording and reporting adverse events

10.3.1 Definitions

- **Adverse event (AE)**

Any untoward medical occurrence in a subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with this treatment.

- **Serious adverse event (SAE)**

Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.

Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.

- **Adverse reaction (AR)**

A response to a medicinal product which is noxious and unintended and which occurs at doses normally used in man for the prophylaxis, diagnosis or therapy of disease or for the restoration, correction or modification of physiological function. This definition also covers medication errors and uses not provided for in the protocol, including misuse or abuse of the product. In determining whether an adverse event is an adverse reaction, consideration shall be given to whether there is a reasonable possibility of establishing a causal relationship between the event and the investigational medicinal product based on an analysis of available evidence.

- **Suspected Unexpected serious adverse reaction (SUSAR)**

A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

- **Important Medical Event**

Adverse event that may not be immediately life-threatening or result in death or hospitalisation but may jeopardise the subject or require medical or surgical intervention to prevent one of the outcomes listed under the definition of Serious Adverse Event. Examples of such events include invasive or malignant cancers, intensive treatment (in an emergency room or at home) for allergic bronchospasm, blood dyscrasias or convulsions that do not result in hospitalisation, or the development of drug dependency or abuse, overdose or misuse.

- **Unexpected event**

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An unexpected event which affect the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42 of Regulation (EU) No 536/2014 of the European parliament. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator shall take appropriate urgent safety measures to protect the subjects.
The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

- **Serious breach**

A 'serious breach' means a breach likely to affect to a significant degree the safety and rights of a subject or the reliability and robustness of the data generated in the clinical trial.

NB: The sponsor is responsible for determining whether the situation corresponds to a serious breach. The sponsor must record and document the serious breach, analyse its fundamental causes, record the information in the CTIS within the time scale described in the Clinical Trial Regulation and take corrective and preventive measures where relevant. In the event of a serious breach likely to affect the benefit/risk ratio of the clinical trial, the notification in the CTIS may be accompanied by other notifications such as those relating to an unexpected event with an impact on the benefit/risk ratio of the clinical trial, urgent safety measures or a declaration of an unexpected serious adverse reaction (SUSAR) for example.

- **Medication error**

The unintentional omission or commission of an action during a treatment process involving a medicinal product that may cause a risk (potential error) or an adverse reaction in the patient.
A medication error may be:
a. Proven and cause an adverse reaction (error with adverse drug reaction) or not cause an adverse reaction (error without harm)
b. Intercepted before administration to the patient (intercepted error)
c. A potential error exists if an observation identifies a potential danger to the patient.

- **Abuse of a medicinal product**

Persistent or sporadic, intentional excessive use of one or more medicinal product which is accompanied by harmful physical or psychological effects.

- **Misuse**

Intentional use that does not comply with the clinical trial framework for an inappropriate medical purpose.

- **Overdose**

Administration of a quantity of a medicinal or other product, given per administration or cumulatively, which is above the maximum recommended dose according to the summary of product characteristics. In practice this is an exposure resulting in high plasma concentrations. It may involve the excessive taking (intentional or accidental) of a medicinal product.

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10.3.2. The role of the investigator

The investigator must **assess the seriousness of each adverse event** and record all serious and non-serious adverse events, and all special circumstances (medication errors, in utero exposure / pregnancy / breastfeeding, uses outside what is foreseen in the protocol, including misuse and abuse of the product) in the case report form (eCRF).

For all serious adverse events, the investigator **must also notify the SAE to the safety department** without undue delay but not later than within 24 hours of obtaining knowledge of the events, except SAE not requiring a notification without delay described in section 10.3.2.2.2. "Serious adverse events that do not require the investigator to notify the sponsor without delay".

Commenté [SE101]: Vérifier n° paragraphe avant soumission

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the intensity** of the adverse events:

- by using the Common Terminology Criteria for Adverse Events [National Cancer Institute] rating scale for adverse events appended to the protocol
- by using tables below (extract of Toxicity Grading Scale for Healthy Adult and Adolescent Volunteers Enrolled in Preventive Vaccine Clinical Trials) for the local and general solicited adverse events.

Commenté [LGC102]: Prévoir les 2 échelles dans les formulaires de notification svp

Local Reaction to Injectable Product	**Mild (Grade 1)**	**Moderate(Grade 2)**	**Severe (Grade 3)**	**Potentially Life Threatening (Grade 4)**
Pain	Does not interfere with activity	Repeated use of non-narcotic pain reliever > 24 hours or interferes with activity	Any use of narcotic pain reliever or prevents daily activity	Emergency room (ER) visit or hospitalization
Erythema/Redness *	2.5 – 5 cm	5.1 – 10 cm	> 10 cm	Necrosis or exfoliative dermatitis
Induration/Swelling **	2.5 – 5 cm and does not interfere with activity	5.1 – 10 cm or interferes with activity	> 10 cm or prevents daily activity	Necrosis

* In addition to grading the measured local reaction at the greatest single diameter, the measurement should be recorded as a continuous variable.

** Induration/Swelling should be evaluated and graded using the functional scale as well as the actual measurement.

Note: In the event of occurrence of prurit, "Illness or clinical adverse event" scale will be used (see last line of table below).

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Systemic (General)	Mild (Grade 1)	Moderate(Grade 2)	Severe (Grade 3)	Potentially Life Threatening (Grade 4)
Nausea/vomiting	No interference with activity or 1 – 2 episodes/24 hours	Some interference with activity or > 2 episodes/24 hours	Prevents daily activity, requires outpatient IV hydration	ER visit or hospitalization for hypotensive shock
Diarrhea	2 – 3 loose stools or < 400 gms/24 hours	4 – 5 stools or 400 – 800 gms/24 hours	6 or more watery stools or > 800gms/24 hours or requires outpatient IV hydration	ER visit or hospitalization
Headache	No interference with activity	Repeated use of non-narcotic pain reliever > 24 hours or some interference with activity	Significant; any use of narcotic pain reliever or prevents daily activity	ER visit or hospitalization
Fatigue	No interference with activity	Some interference with activity	Significant; prevents daily activity	ER visit or hospitalization
Myalgia	No interference with activity	Some interference with activity	Significant; prevents daily activity	ER visit or hospitalization
Fever (°C) ** (°F) **	38.0 – 38.4 100.4 – 101.1	38.5 – 38.9 101.2 – 102.0	39.0 – 40 102.1 – 104	> 40 > 104
Illness or clinical adverse event (as defined according to applicable regulations)	No interference with activity	Some interference with activity not requiring medical intervention	Prevents daily activity and requires medical intervention	ER visit or hospitalization

Note: In the event of occurrence of abdominal pain, chills or malaise, “Illness or clinical adverse event” scale will be used (see line above).

The investigator must **assess the causal relationship** between serious adverse events and:

- the investigational medicinal product (IMP),
- and/or interventions/procedures added by the study (more elevated blood volume taken) and/or auxiliary medicinal product
- and/or interaction between auxiliary medicinal product and the investigational medicinal product.

Commenté [LGC103]: S’il n’y a pas de prélèvements complémentaires et qu’il s’agit uniquement de tubes complémentaires, pas de vigilance à réaliser

Merci de clarifier svp

In the absence of information on causality from the investigator submitting the report, the sponsor consults with the investigator and encourages him or her to provide an opinion on this matter. The sponsor shall not downplay the significance of the causal relationship established by the investigator in his or her assessment. If the sponsor disagrees with the investigator’s assessment of the causal relationship, both the investigator’s and the sponsor’s opinions shall be included in the report.

The method to be used by the investigator is based on 2 causality terms (EudraVigilance Clinical Trial Module (EVCTM) method):

- Related
- Not related

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10.3.2.1. Serious adverse events that require the investigator to notify the sponsor without delay

The investigator notifies the sponsor without undue delay but not later than within 24 hours on the day the investigator becomes aware of any serious adverse event which occurs during a study that meets the description in article 41 of Regulation (EU) N°536/2014, with the exception of any event which is listed in the protocol and, if applicable, in the investigator's brochure as not requiring notification.

If the investigator becomes aware of a serious adverse event, suspected to be causally related to the investigational medicinal product, occurring after the end of the clinical trial in a participant whom they have treated, they shall inform the sponsor without undue delay.

A serious adverse event is any untoward medical occurrence that:

1- results in death
2- is life-threatening to the participant enrolled in the study
3- requires hospitalisation or prolongation of existing hospitalisation
4- results in persistent or significant disability/incapacity
5- is a congenital anomaly/birth defect
6- important medical event*

***** *Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.*

Moreover, special circumstances require the investigator to notify the sponsor without delay are:

- medication errors,
- in utero exposure / pregnancy / breastfeeding,
- uses outside what is foreseen in the protocol, including misuse and abuse of the product.

All pregnancies must be reported to the safety department by the investigators within 24 hours on the day the investigator becomes aware of it.

Only special circumstances associated to an SAE must be reported to the safety department by the investigators and must be recorded in the "SAE" module of the CRF.

Special circumstances not associated to an SAE must be reported to the CRF in the "AE" module by the investigators and notify as a deviation to the protocol by the clinical trial unit.

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~~**Special circumstances not associated to an SAE must be reported to the CRF by the investigators and notify as a deviation to the protocol by the clinical trial unit.**~~

10.3.2.2. Specific features of the protocol

10.3.2.2.1. Other events that require the investigator to notify without delay the sponsor

- Adverse events particularly followed by the sponsor for the safety and assessment

These events must be notified by the investigator to the sponsor without delay on the day the investigator becomes aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).

These adverse events particularly followed are any DLT.

For these adverse events particularly followed by the sponsor, please, check the "important medical event" box as a seriousness criterion in the SAE notification form.

- **In utero exposure**

The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs during the study even if it is not associated with an adverse event.

Notification is done using a specific form appended to the protocol.

- **Exposure while breastfeeding**

Exposure while breastfeeding occurs if an infant or a child may have been exposed to a medicinal product via the breast milk of a mother being treated with an investigational medicinal product.

Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware of the exposure while breastfeeding.

10.3.2.2.2. Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are only recorded in the case report forms.

An extraction of all these SAE from the e-CRF will be carried out by Clinical Trial Unit (including randomisation group if applicable, seriousness criteria, investigator's assessment and SAE grade if applicable) and sent by email to the Safety Department once per year during the preparation of annual safety report. These extractions will be sent to the DRCI safety department at the following address: expertisecsi.drc@aphp.fr

- *Normal and natural course of the condition*:

Commenté [CoD104]: Il ne faut pas parler des DLT ici avec notification dès connaissance

Commenté [SE105R104]: Effectivement, cela a été ajouté.

a mis en forme : Couleur de police : Automatique

a mis en forme : Police :(Par défaut) Arial, 11 pt, Non Italique, Anglais (Royaume-Uni)

a mis en forme : Police :(Par défaut) Arial, 11 pt, Anglais (Royaume-Uni)

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This includes in particular:

- Relapse
- Occurrence of a 2nd leukaemia
- Bacterial or non-bacterial infection (fungal or viral agent) requiring specific anti-infectious treatment in hospital (e.g., chickenpox, EVB replication requiring the use of rituximab, etc.).
- Grade 3-4 haematological toxicities or hepatic cytolysis occurring in the context of chemotherapy administered for AML (see also paragraph below: These serious adverse events (except those with a fatal outcome) are only recorded in the observation book.
- VZV infections meeting a severity criterion (except those with a fatal outcome) will not be notified immediately to the sponsor and are only recorded in the observation book.
- Hospitalization for the natural evolution of the pathology: for example, relapse, occurrence of a 2nd leukaemia, bacterial or non-bacterial infection.

  - *Other circumstances that do not require the investigator to notify the sponsor without delay*

- Hospitalization for medical or surgical treatment scheduled before the research
- Hospitalization for a pre-existing pathology
- Hospitalization for social or administrative reasons
- Medication errors or uses outside what is foreseen in the protocol, including misuse and abuse of the product without SAE and not in relationship with a serious.

10.3.2.3. Period during which SAEs must be notified without delay by the investigator to the sponsor

The investigator must notify the sponsor without delay of any serious adverse events as defined in the corresponding section:

- From the date on which the participant signs the consent form to the study
- Throughout the whole follow-up period required for the trial after the end of the participant's treatment with the investigational medicinal product.
- Indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear long after exposure to the medication, such as cancers or congenital abnormalities) *.

*NB: In this last case, the investigator **does not have to collect indefinitely in the case report form (eCRF)** all **SAEs possibly related** to the clinical trial, **but must notify them**, to the sponsor, **as soon as he/she becomes aware of them, by the SAE notification form or by email or by fax (as described below)**.

10.3.2.4. Procedures and deadlines for notifying the sponsor

The initial notification of an SAE must be provided in a written report signed by the investigator using a SAE notification form specific to the study and intended for this purpose (in the case report form).

Each item in this form must be completed by the investigator so that the sponsor can carry out the appropriate analysis.

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The initial notification to the sponsor of a serious adverse event must be quickly followed by an additional detailed written report (or reports) so that the case outcome may be monitored by the Safety Department or to provide further information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful (medical reports, laboratory test results, results from additional examinations, etc.). These documents must be non-identifying. In addition, the documents must include the following: study acronym and number and participant's reference number.

Commenté [CoD106]: Revoir mode de transmission par rapport à la MR001

Commenté [SE107R106]: Pas de problème

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the participant has left the study.

Since the observation log is electronic and connected to the Safety Department's database, information will be transmitted exclusively through this channel.

The initial notification, follow-up reports for a vigilance case involving one or more serious adverse events (SAEs), and any other documents will be transmitted to the sponsor, represented by the Safety Department, electronically via the interface between the electronic case report form and the vigilance database.

To do this, the investigator completes the notification form for a vigilance case involving one or more serious adverse events in the e-CRF, saves it, and then clicks the "Notify" button.

By performing this action, the investigator confirms and validates the transmitted data.

Procedure to follow in the event of a malfunction:

If the electronic case report form (e-CRF) is unavailable or there is a connection failure between the e-CRF and the vigilance database, the initial report, follow-up reports for a vigilance case, and any other documents must be sent to the sponsor, represented by the Safety Department, via email (eig-vigilance.drc@aphp.fr).

If the e-CRF is operational but the connection gateway to the vigilance database is malfunctioning:
- The investigator completes the safety case report form in the e-CRF, saves it, prints it, signs it, and then sends it via email (eig-vigilance.drc@aphp.fr).
- As soon as the connection is restored, the investigator submits the adverse event report form, which was completed and saved in the e-CRF, via the connection gateway.

If the e-CRF is unavailable:
- The investigator completes, signs, and sends the SAE notification form to the Safety Department via email (eig-vigilance.drc@aphp.fr).

If the participant experiences multiple serious adverse events, one serious adverse event notification form must be completed for each event.

- As soon as access is restored, the investigator corrects the record by completing the pharmacovigilance case notification form, which covers one or more serious adverse events, in the e-CRF and then submits it via the connection gateway.

When sending by email, please:
- Adopt a standardized nomming of the email subject in the following form:

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<< Objet : YYYYYY_XXXXXX_jjmmaaaa (where YYYYYY is the research code, XXXXXX is the research acronym, and ddmmyyyy is the date of submission). >>

- Send a SAE initial notification form and/or a follow-up report concerning a single participant for a given SAE, attachment may contain one (or more) document(s) (follow-up and hospitalization reports, for example).
The total size of the email must be less than 8 MB, otherwise please send several e-mails.

- Ensure that all documents transmitted (e.g. hospital reports) are anonymized and identified with the participant's identification number.

~~When sending the email, please:~~
~~- Adopt a standardized nomming of the email subject in the following form:~~
~~<< Objet : YYYYYY_XXXXXX_jjmmaaaa (avec YYYYYY : code de la recherche, XXXXXX : acronyme de la recherche et jjmmaaaa : date de transmission). >>~~

a mis en forme : Anglais (États-Unis)

~~- Send a SAE initial notification form and/or a follow-up report concerning a single participant for a given SAE, attachment may contain one (or more) document(s) (follow-up and hospitalization reports, for example).~~
~~The total size of the email must be less than 8 MB, otherwise please send several e-mails.~~

~~- Ensure that all documents transmitted (e.g. hospital reports) are pseudonymized and identified with the participant's identification number.~~
~~For studies which use e-CRFs:~~

The investigator must respond to all requests from the sponsor for additional information (these new information's must be also recorded in the e-CRF).

~~- if it is not possible to connect to the e-CRF, the investigator should complete, sign, and send the SAE notification form to the Safety Department. As soon as the connection is restored, the SAE notification form in the e-CRF must be duly completed.~~

The investigator must respond to all requests from the sponsor for additional information (these new information's must be also recorded in the e-CRF).

For all questions relating to the notification of an adverse event, the Safety Department can be contacted via email: vigilance.drc@aphp.fr.

For cases of *in utero* exposure, the investigator will complete the Notification and Follow-up form for a pregnancy occurring during participation in a study".
The investigator must monitor the pregnant woman throughout her pregnancy or until the pregnancy is terminated, and must notify the sponsor of the outcome of the pregnancy using this form.
If the outcome of the pregnancy falls within the definition of a serious adverse event (miscarriage, pregnancy termination, foetal death, congenital abnormality, etc.), the investigator must follow the procedure for reporting SAEs.
The initial pregnancy notification, the SAE follow-up reports, and any other documents will be sent to the sponsor according to the same procedures specified herein.

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10.3.3. Role of the sponsor

The sponsor, represented by its Safety Department, shall continuously, throughout the trial, assess the safety of each investigational medicinal product throughout the study.

10.3.3.1. Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all the adverse events reported
- the **causal relationship** between these events and the investigational medicinal product and/or the auxiliary medicinal product and/or interaction between the investigational medicinal product and the auxiliary medicinal product and any concomitant treatments,

  All serious adverse events which the investigator and/or the sponsor reasonably believe may have a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.
- the **expected or unexpected nature** of the serious adverse reactions

  Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.

  The sponsor, represented by its Safety Department, assesses the expected/unexpected nature of a serious adverse reaction based on the information described below**.**

❖ For serious adverse events likely to be related to the investigational medicinal product:

❖ For serious adverse events likely to be related to the investigational medicinal product~~(s)~~:

Refer to the section 4.8 “Undesirable effects” of the SmPC for Shingrix® uploaded to CTIS platform and enclosed in the investigator study file.

❖ For serious adverse events that may be related to the auxiliary medicinal product: Refer to the SmPC for “EMLAPATCH 5%”, mentioned in the cover letter and enclosed in the investigator study file.

~~Expected vaccine reactions are local (pain, swelling, redness, pruritus at the injection site) and general symptoms (chills, fever, fatigue, malaise, myalgia, headache, abdominal pain, diarrhoea, nausea and vomiting).~~

❖ For serious adverse events that may be related to the auxiliary medicinal product: Refer to the SmPC for “EMLAPATCH 5%”, mentioned in the cover letter and enclosed in the investigator study file.

Commenté [LGC109]: A garder uniquement si prélèvements sanguins ajoutés

Commenté [CA111]: Les RCP des MA ne sont pas enregistrés dans CTIS

Commenté [CA111]: Les RCP des MA ne sont pas enregistrés dans CTIS

a mis en forme : Police :Gras, Couleur de police : Couleur personnalisée(RVB(31;73;125))

a mis en forme : Police :Gras, Couleur de police : Couleur personnalisée(RVB(31;73;125))

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❖ The serious adverse events potentially related to the interventions, procedures or examinations specific to the study (blood samples) are: hematoma, pain, vagal malaise, mild bleeding.

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs) via Eudravigilance, within the regulatory time frame, to the competent authority:

- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction;
- in the case of non-fatal or non-life-threatening suspected unexpected serious adverse reactions, not later than 15 days after the sponsor became aware of the reaction;
- in the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction being fatal or life-threatening. If the initial report of a SUSAR was initially considered to be non-fatal or non-life-threatening but which turns out to be fatal or life-threatening is incomplete, a follow-up report shall be made as soon as possible, but within a maximum of 7 days of first knowledge of the reaction being fatal or life threatening. The sponsor shall submit a completed report within an additional 8 days
- in the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and, in any events, not later than eight days after the initial report
- in cases where a SUSAR turns out to be fatal or life-threatening, whereas initially it was considered as non-fatal or not life-threatening, if the initial report has not yet been submitted, a combined report shall be created
- all additional, relevant information must be declared by the sponsor in the form of follow-up reports within a period of 15 days starting from when the sponsor had this information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

10.3.3.2. Analysis and declaration of other safety data

This means any new information that prompts a reassessment of the risk/benefit ratio of the trial or of the product under investigation, or which could be sufficient to consider changes to the use of the product, the trial procedures or trial documentation, or the suspension, cancellation or amendment of the research trial or of other similar trials. For trials involving the first administration or use of a health product in healthy volunteers: any serious adverse reaction.

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The sponsor will report in CTIS platform without delay upon knowledge of any emerging safety issue and, if applicable, describe what measures have been taken.

The unexpected event shall be notified without undue delay but no later than 15 days from the date the sponsor became aware of this event.

The urgent safety measures shall be notified without undue delay but no later than 7 days from the measures have been taken.

Following the initial declaration of any emerging safety issue, the sponsor will address to competent authorities any additional relevant information about the emerging safety issue in a follow-up report.

10.3.3.3. Annual safety report

Once a year for the duration of the clinical study, the sponsor must produce an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- a safety analysis for the research participants,
- a description of the patients included in the study (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- summary tables of all the serious adverse events that have occurred since the start of the study.

The sponsor produces one annual safety report (Development Safety Update Report - DSUR) for one clinical trial.

The report must be submitted in CTIS no later than 60 days after the anniversary of the date on which the competent authority authorised the trial.

The final annual safety report must be submitted in CTIS no later than 60 days after the end date of the clinical trial.

The end date of the clinical trial is defined as the date of the last visit of the last subject.

10.3.4. Data Safety Monitoring Board (DSMB)

A DSMB will be established for this trial. The DSMB must hold its first meeting before the first participant is enrolled and ideally before the protocol is submitted to the competent authority and the Ethics Committee).

All missions including monitoring of safety data, as well as the specific operating procedures of the DSMB are described in the study's DSMB charter.

The DSMB members are:

- Pr. Stéphane Ducassou (Hematology and Oncology, CHU de Bordeaux)

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- Madame Sarah Zohar (Biostatistic, Unité de Recherche Inserm (U1138)
- Dr Marion Caseris (Infectious disease specialist, Hôpital Robert Debré, APHP, Paris)

The DSMB will operate in accordance with the sponsor's procedures. The DSMB works in an advisory capacity only and the sponsor retains all decision-making authority.

11. DATA MANAGEMENT

11.1. Data collection procedures

Clinical and biological data will be collected in an e-CRF (CleanWeb®).
All data could be then entered in the Doremy database.

11.2. Identification of data recorded directly in the CRFs which will be considered as source data

No data recorded directly in the e-CRF will be considered as source data.

11.3. Right to access data and source documents

11.3.1. Data access

In accordance with GCPs and appendix 1 of the European Regulation N°536-2014:

- The sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures or inspections by the competent authority
- The sponsor declares that investigators and participating institution will ensure the persons in charge of monitoring, quality control and auditing or inspecting the clinical trial have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force.

11.3.2. Source documents

Source documents are defined as any original document or item that can prove the existence or accuracy of a data or a fact recorded during the study. These documents will be kept in accordance with the regulations in force by the investigator or by the hospital in the case of a hospital medical file.
In the context of the study, list the source documents are medical file, original biological examination results, diary card.

11.3.3. Data confidentiality

The persons responsible for the quality control of clinical studies will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study participants and in particular their identity and the results obtained.

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These persons, as well as the investigators themselves, are bound by professional secrecy.

During and after the clinical trial, all data collected concerning the participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be rendered non-identifying.

Under no circumstances shall the names and addresses of the participants involved be shown.

Only the participant's initials will be recorded, accompanied by an encoded number specific to the study indicating the order of enrolment.

The sponsor will ensure that each participant and his/her holder(s) of legal authority has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4. Data processing and storage of research documents and data

11.4.1. Identification of the data processing manager and location(s)

The statistical analysis will be carried out in the biostatistics and medical information department (SBIM) at Saint-Louis Hospital under the responsibility of Professor Chevret.

11.4.2. Data entry

Non-identifying data will be entered electronically via a web browser.

11.4.3. Data ownership

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12. STATISTICAL ASPECTS

12.1. Description of statistical methods to be used including the timetable for the planned interim analyses

12.1.2 Phase 1

The analysis of the DLT endpoint will use a Bayesian framework.

Sequential analyses of data will be performed after each cohort of 2 patients, following the Continual Reassessment Method (CRM) dose-finding clinical trial design. It is a model-based design, where the dose–limiting toxicity relationship is modelled through a cumulative function corresponding to a one-parameter power model, the parameter of which is sequentially estimated from the data.

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12.1.2 Phase 2

Control of biases

Randomization (that allows controlling selection bias) will be centrally performed, to ensure allocation concealment. It will be further stratified to ensure the balance across randomized groups of the main prognostic factor (that is, VZV+/-) avoiding confounding bias.

Estimand

We will focus on the Treatment Policy Estimand, based on:

- The Intent-to-Treat/vaccine (ITT) population, defined as all subjects who are included, independent of whether they received study treatment or not. Every patient included in the study will be taken into account in the final intent-to-vaccine analysis.
- The main outcome measure is the prevalence of an immunologic response at month-2 following randomisation.
- The measure of treatment effect will be both a relative measure, based on relative risk (RR), and absolute measures, based on the difference in response rates between the randomized groups.
- All the intercurrent events that occur after randomisation (including the lack of second injection in the experimental group) will be ignored in the analysis

Baseline description

A descriptive analysis of the characteristics of patients and controls will be conducted.

Qualitative data will be described in frequency and percentage. Quantitative data will be describedusing median and inter-quartile range (IQR).

Primary endpoint

Analysis will be performed once the expected number of patients to control the power has been reached, that is 76 (see section 12.2).

The probabilities of response will be estimated and compared across groups on the basis of the exact Fisher test. Measure of treatment effect will be both relative, based on the relative risk (RR) of response, and absolute measures, using difference in response rates, as measured at 2-month, both with 95% exact confidence intervals.

Secondary endpoints

All proportions will be analysed similarly as the main endpoint.

Cumulative incidence of VZV infection will be estimated in a competing-risks setting, where deaths prior to infection compete with the event of interest. The cause-specific hazard Cox model will be used to estimate the hazard ratio (HR) of infection, from which will be derived the relative vaccine efficacy (VE).

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12.2. Calculation hypotheses for the number of participants required and the result

Phase 1

Commenté [FAL112]: faut il marquer que du coup on va inclure 12 patients dans la phase 1?

The sample size calculation was based on empirical approximation for the CRM using the power (or empiric) dose-toxicity function, F (d, β) = dexp(β), where β has a normal prior with mean 0 and variance 1.34, and the starting dose is the median level (22).

The skeleton and prior variance of the dose-toxicity model parameter will be further calibrated by simulation (23), to optimize the probability of correct selection (PCS), accounting for the trial design and additional safety rules, and detailed in the SAP before trial initiation.

The sample size will likely allow convergence of dose assignments on the estimated MTD and result in an expanded sub-sample of patients assigned to the estimated MTD.

Phase 2- Initial calculation of a provisional sample size. Sample size calculation used the primary endpoint, that is, the prevalence at month 2 of patients with immunologic response, stratifying the analysis independently on each stratum defined according to the type of VZV status, while controlling on each stratum for the 2.5% two-sided type I error rate (due to the 2 populations) and a statistical power at 95%, to demonstrate a benefit.

(i) in VZV- children, we hypothesized an increase in the immunologic response rate from 5% to 50%; this required the inclusion of two groups of 20 subjects.

(ii) in VZV+ children, we hypothesized an increase in the immunologic response rate from 5% to 60%; thus, 2x13 subjects had to be included.

These hypotheses relied on the absence of immunologic response in absence of any vaccine, thus setting at 5% the observed rate in the control groups.

This calculation was based on the results of the Dagnew et al. study for VZV+ subjects and the Lhuillier et al. study for VZV- subjects.

In the study by Dagnew et al. (17) which investigated the immunogenicity of Shingrix® vaccine in VZV+ adults with haematological malignancies, the proportion of patients with a vaccine response (assessed by the level of anti-gE IgG) after 2 doses of vaccine was 80.4% (95% CI: 73.1 - 86.5%) in the vaccinated group versus 0.8% (0.0 - 4.2%) in the placebo group. We cannot transpose these results to our study because half the patients were no longer taking IS at the time of vaccination. Indeed, our patients will all be undergoing chemotherapy at the time of vaccination and will therefore probably be more immunocompromised than the population in this study. We have therefore chosen an estimated immunogenicity rate of 60% in VZV+ subjects, which is slightly lower.

In the study by Lhuillier et al. (12), the only study to assess the immunogenicity of Shingrix® in 31 VZV- subjects (adult solid organ transplant recipients), seroconversion was observed in only 55% of patients after 2 doses of vaccine.

As the population in our study is probably more immunosuppressed (ongoing chemotherapy), we have chosen an estimated immunogenicity rate of 50%, slightly lower than that observed for VZV- children.

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Thus, a total of n0=40 VZV- and 26 VZV+, i.e., 66 children with LA is scheduled to be enrolled.

Re-estimation of the size of the sample size to include in each arm, $\widehat{N}$, after observation of the responses of a sample of size n1=n0/2=33 first included patients.

This re-estimation will be based on the estimated overall proportion of immunologic response, obtained by substituting this rate by its estimate $p = \frac{x_C + x_T}{n_1}$ where $x_C + x_T$ is the total number of observed responses in these n1 patients, while other quantities are unchanged (24).

The re-estimated sample size will then be expressed as a function of the initial sample size n0: $\widehat{N} = n_0 \frac{p(1-p)}{\pi(1-\pi)}$. This formula shows that such a computation is independent from the type I and II error rates (25).

Thus, the sample size to be further included in each arm is given by n2=max(n1, $\widehat{N}$)-n1(« unrestricted design ») (26).

Of note, this sample size may be limited by the independent trial committee due to budget or recruitment constraints (25).

12.3. Anticipated level of statistical significance

The type I error rate was 5%, divided in 2.5% in each of the two populations (VZV- and VZV+); this would allow to test the primary outcome measure in each subset of children.

12.4. Statistical criteria for termination of the study

One interim analysis is planned after the inclusion and observation of response of 38 patients. Even if there is a theoretical possibility of inflated type I error rate using this re-estimation process based on the reevaluation of the overall proportion of response, blinded to treatment arm, this approach has been shown asymptotically valid without any substantial inflation of the type I error rate I (27; 28). Nevertheless, as a conservative approach, the Haybittle-Peto approach will be used, with a p-value of 0.002 or less required to conclude against the null hypothesis.

12.5. Method for taking into account missing, unused or invalid data

No missing data is expected in this randomized trial that targets children with acute leukaemia. The only cause of missing data could be the study dropout with refusal to use further data; in this case, imputation of missing data will be performed, using method adapted to the type of missing mechanism, and with sensitivity analyses. All will be defined in the statistical analysis plan (SAP) before study enrolment.

12.6. Management of modifications made to the analysis plan for the initial strategy

All modifications will be amended in the SAP.

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12.7. Choice of individuals to be included in the analyses

All randomized children will be included in the analyses, whichever they were administered the experimental or not, and analysed in the group they were allocated to.

13. QUALITY CONTROL AND ASSURANCE

Every clinical trial sponsored by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored clinical trial.

13.1. General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the study. The sponsor must implement a quality assurance system to best monitor the implementation of the study in the investigation centres.

For this purpose, the sponsor shall assign Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study locations, after having carried out the initial visits.

The purpose of monitoring the study, as defined in Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1. Strategy for centre opening

The strategy for opening the centres established for this study is determined using the appropriate monitoring plan.

13.1.2. Scope of centre monitoring

In the case of this **D** risk study the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact, resulting in a study monitoring level to be implemented: level **D**.

13.2. Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical

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Research and Innovation Department) and in accordance with Good Clinical Practices as well as the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits carried out by the Clinical Research Associate. During these visits, the following elements will be reviewed depending on the monitoring level:

- written consent of both holder(s) of legal authority and of participant (if possible)
- compliance with the study protocol and with the procedures defined therein
- quality of the data collected in the case report form: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used.

13.3. Case report forms

All information required by the protocol must be entered in the electronic case report forms (eCRF, CleanWeb®). The data must be collected as and when they are obtained, and clearly recorded in these case report forms. Any missing data must be coded. This study will be associated with DoReMy database, sponsored by AP-HP.

Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given a document offering guidance on using this tool.

When the investigators complete the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, the data are immediately verified as they are entered, thanks to consistency checks. To this end, the investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment.

The investigator must archive a copy of the authenticated document that was issued to the sponsor.

13.4. Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study – with the protocol, standard operating procedures, good clinical practices or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

The sponsor shall notify the Member States concerned about a serious breach of this Regulation or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than seven days of becoming aware of that breach.

13.5. Audits/inspections

The investigators agree to consent to the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and

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reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the grounds of medical secrecy.

An audit can be carried out at any time by individuals appointed by the sponsor and independent of those responsible for the research. The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the study agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results, including the storage of the data used or produced as part of the study.

13.6. Principal Investigator's commitment to assume responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal *curriculum vitæ*, signed and dated less than one year, with his/her RPPS number (*Répertoire Partagé des Professionnels de Santé*, Collective Database of Health Professionals, for France). The CV must include any previous involvement in clinical research and related training.

Any conditions, such as economic interests and institutional affiliations that might influence the impartiality of the investigators shall be presented.

Each investigator will commit to comply with legislation and to conduct the study in line with GCP, in accordance with the Declaration of Helsinki. Each investigator will give the sponsor's representative a GCP certificate dated fewer than three years ago.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.

The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

13.7. Suitability of the facilities

Before starting the study, each clinical trial sites will give the sponsor's representative a duly justified written statement on the suitability adapted to the nature and use of the investigational medicinal product and including a description of the suitability of facilities, equipment, human resources and description of expertise.

13.8. Pharmacist's commitment of responsibility

Commenté [PC113]: A valider par AGEPS

The AGEPS's pharmacist is responsible for purchasing the vaccines, storing them, sending them to the centre pharmacies and destroying any remaining doses. He/she will also be responsible for the traceability of these various stages.

The pharmacy at each site will be responsible for receiving Shingrix® vaccines, storing them and dispensing them in the departments, destroying any remaining doses as well as ensuring that each of these stages can be traced.

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14. ETHICAL AND LEGAL CONSIDERATIONS

14.1. Methods for informing research participants and obtaining their consent

Information of the holder(s) of legal authority and its/their consent in the case of a study protocol involving a minor

In accordance with Article 32 of the European regulation, when a clinical trial is conducted on a non-emancipated minor, consent must be given by the holder(s) of its legal representative(s).

During a standard of care visit (screening visit), oral explanation and Information note is given to the minor and the holder(s) of parental authority by the investigator, or his/her representative.

A reflection period of 3 days to 3 months is given between the time when the interested persons are informed and when they sign the consent form (Baseline visit).

The freely-given written informed consent of the holder(s) of legal authority is obtained by the investigator, or by a physician representing the investigator, before definitive inclusion of the minor in the study (baseline visit, M0).

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Information for minors participating in the research

Minors receive the information specified in Article 29.2 of the European regulation, appropriate to their level of understanding, both from the investigator and from the holder(s) of legal authority.

In this study, minors aged between 1 and 17 will be included. Three models of information's sheet will be provided according to age of the participant (under 7; 7 to 11 and over 11).

Minor's personal endorsement is sought regarding their participation in the study involving human participants. In any cases, the investigator cannot override their refusal or the revocation of their acceptance.

One copy of the signed and dated consent form is given to the holder(s) of legal authority. The principal investigator or a physician representing him/her will keep one copy.

At the end of the study, one copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.

Information recorded in the minor's medical file

The investigator will record the minor's participation in the clinical study in the minor's medical file, along with the procedure for informing and obtaining consent from the holder(s) of legal authority as well as the procedure for informing the minor and a record of the minor's non-rejection to take part.

Special circumstances: the minor reaches the age of majority during his or her participation in the study.

According to article 32.3 of the European regulation, Minors who reach the age of majority during their participation in the study will be given new, relevant information at that time. After they have been given this information, they will be asked to confirm their consent.

14.2. Prohibition from participating in another clinical study or exclusion period set after the study

No exclusion period of participation after the participant has finished this study is defined in the context of this research.

The participant may enrol in another interventional study (except vaccine clinical trial) and non-interventional studies for the duration of his or her participation only with consulting with the physician monitoring him or her in the context of the study.

14.3. Compensation for participants

No compensation will be paid.

14.4. Authorisation for the research location

Patients will be included in the study in oncology/ paediatric units. The interventions performed on these participants are acts usually conducted at these units. We therefore consider that the research location does not require specific authorisation.

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14.5. Legal obligations

14.5.1. Role of the sponsor

Assistance Publique - Hôpitaux de Paris (AP-HP) is the sponsor of this study and, by delegation, the DRCI (Clinical Research and Innovation Department) carries out the study's missions in accordance with regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and all national Laws. Assistance Publique - Hôpitaux de Paris reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.5.2. Request for authorisation

Prior to starting the study, AP-HP, as sponsor, must obtain authorisation from the Competent Authority and the Research Committee for this clinical trial on medicinal product for human use within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.5.3. Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended "Informatique et Libertés" law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

This research is governed by the CNIL (French Data Protection Agency) "Reference Methodology for processing personal data used within the scope of health research" (amended MR-001). AP-HP, as sponsor of the research, has signed a declaration of compliance with this "Reference Methodology".

Commenté [PC114]: A confirmer pour la teleconsultation (en attente retour CNIL)

Commenté [PC115R114]: OK MR001, il faut que le logiciel soit installé sur un poste de travail et non une solution en ligne

Commenté [CoD116]: Je crois que c'est bon, il faut par contre pouvoir s'assurer de l'identification du patient par le medecin donc caméra.

Commenté [SE117R116]: On a eu le retour donc c'est bon.

Commenté [SE118R116]: @Christine : Suite à la mise à jour de la MR001, des modifications de ce point doivent-elles être rédigées ?

Commenté [PC119R116]: Pas de modification

14.5.4. Start of the Clinical Trial

The sponsor shall notify each Member State concerned of the start of a clinical trial and the start of the recruitment through the EU portal. That notification shall be made within 15 days from the start of the clinical trial.

14.5.5. Amendments to the Clinical Trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain approval from the Ethics Committee) and authorisation from the competent authority within the scope of their respective authorities, before the amendment can be implemented.
The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.5.6. End of the Clinical Trial

The end of the Clinical Trial corresponds to the end of the participation of the last person who participate to the Clinical Trial.

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The end of the recruitment of subjects for the clinical trial and the end of the clinical trial should be notified in CTIS. That notification shall be made within 15 days from the end of the clinical trial in relation to that Member State.

14.5.7. Summary of the results of the clinical trial

According to article 37.4 of the European regulation n°536/2014, irrespective of the outcome of a clinical trial, within one year from the end of a clinical trial in all Member States concerned, the sponsor shall submit to the CTIS a summary of the results of the clinical trial. This summary shall be accompanied by a summary written in a manner that is understandable to laypersons.

14.5.8. Archiving

Specific documents for a clinical trial on medicinal product for human use will be archived by the investigator and the sponsor for 25 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list) :
  - the successive versions of the protocol (identified by the version number and its date), and any appendices
  - the competent authority authorisations and Research Ethics Committee decisions
  - any correspondence
  - the enrolment list or register
  - the appendices specific to the research
  - final study report
- The data collection documents.

15. FUNDING AND INSURANCE

15.1. Funding sources

PHRC-I (Hospital Funding for Clinical Research).

15.2. Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

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Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with AXA XL through HOWDEN France for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article 76(1) of regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and Article L.1121-10 of the Code de la Santé Publique (French Public Health Code).

Commenté [CoD120]: Toujours eux ?

Commenté [SE121R120]: @Christine : Peux-tu confirmer ? Car le devis a été établi avec eux, non ?

Commenté [PC122R120]: Changement depuis janvier 2026 : AXA XL HOWDEN France → budget a mettre à jour et NIFC

16. PUBLICATION RULES

The author(s) of any publication relating to this study must include the AP-HP among their affiliations and must name the sponsor AP-HP (DRCI) and the source of funding; a copy of the publication must be sent to the sponsor (see below for rules governing affiliation, and naming of the sponsor and funders).

16.1. Mention of AP-HP affiliation for projects sponsored by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is unimportant
- Each of these affiliations must be identified by an address and separated by a semicolon (;)
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France.

16.2. Mention of the sponsor AP-HP (DRCI) in the acknowledgements of the text

The sponsor was Assistance Publique – Hôpitaux de Paris (Direction de la Recherche Clinique et de l'Innovation)".

16.3. Mention of the financial backer in the acknowledgements of the text

The study was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2024 (French Ministry of Health).

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17. BIBLIOGRAPHY

1. Moschovi, M., Adamaki, M. & Vlahopoulos, S. A. Progress in Treatment of Viral Infections in Children with Acute Lymphoblastic Leukemia. *Oncol. Rev.* **10**, 300 (2016).

2. Lin, H.-C. *et al.* Increased risk of herpes zoster in children with cancer. *Medicine* **95**, e4037 (2016).

3. Hansson, E., Forbes, H. J., Langan, S. M., Smeeth, L. & Bhaskaran, K. Herpes zoster risk after 21 specific cancers: population-based case–control study. *Br. J. Cancer* **116**, 1643–1651 (2017).

4. Zawitkowska, J. *et al.* Viral Infection Profile in Children Treated for Acute Lymphoblastic Leukemia—Results of Nationwide Study. *Pathogens* **11**, 1091 (2022).

5. Xiao, P. *et al.* A prospective multicenter study on varicella-zoster virus infection in children with acute lymphoblastic leukemia. *Front. Cell. Infect. Microbiol.* **12**, 981220 (2022).

6. Diniz, L. M. O. *et al.* Study of Complications of Varicella-Zoster Virus Infection in Hospitalized Children at a Reference Hospital for Infectious Disease Treatment. *Hosp. Pediatr.* **8**, 419–425 (2018).

7 Hsiao-Chuan L. *et al.* Increased risk of herpes zoster in children with cancer: A nationwide population-based cohort study. *Medicine* Jul 29;95(30): e4037 (2016).

8. Kim, S.-K. *et al.* Clinical characteristics and outcomes of varicella zoster virus infection in children with hematologic malignancies in the acyclovir era. *Blood Res.* **51**, 249–255 (2016).

9. Wen SY, Liu WL. Epidemiology of pediatric herpes zoster after varicella infection: a population-based study. *Pediatrics*.;135: e565–e571 (2015).

10. Sørensen, G. V., Helgestad, J. & Rosthøj, S. [Herpes zoster-associated morbidity in children undergoing chemotherapy for acute lymphoblastic leukaemia]. *Ugeskr. laeger* **171**, 3350–4 (2009).

11. Haut Conseil de la santé publique. Vaccination des personnes immunodéprimées ou aspléniques: Recommandations. 2e édition - Décembre 2014

12. L'Huillier, A. G. *et al.* Evaluation of Recombinant Herpes Zoster Vaccine for Primary Immunization of Varicella-seronegative Transplant Recipients. *Transplantation* **105**, 2316–2323 (2021).

13. Saied, M. H. et al. Safety, Immunogenicity and Efficacy of the Shingrix Vaccine in Immunocompromised Varicella Zoster Virus Naïve Pediatric Patients. Vaccine 2025; 3: 64: 127708. doi: 10.1016/j.vaccine.2025.127708.

14. HAS • Recommandations vaccinales contre le zona. Place du vaccin Shingrix. 24 février 2024. Mis en ligne le 07 mars 2024.

15. Stadtmauer, E. A. et al. Adjuvanted recombinant zoster vaccine in adult autologous stem cell transplant recipients: polyfunctional immune responses and lessons for clinical practice. *Hum. Vaccines Immunother.* **17**, 4144–4154 (2021).

16. Bastidas, A. *et al.* Effect of Recombinant Zoster Vaccine on Incidence of Herpes Zoster After Autologous Stem Cell Transplantation. *JAMA* **322**, 123–133 (2019).

a mis en forme : Anglais (États-Unis)

a mis en forme : Français (France)

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17. Dagnew, A. F. *et al.* Immunogenicity and safety of the adjuvanted recombinant zoster vaccine in adults with haematological malignancies: a phase 3, randomised, clinical trial and post-hoc efficacy analysis. *Lancet Infect. Dis.* **19**, 988–1000 (2019).

18. Vink, P. *et al.* Immunogenicity and safety of the adjuvanted recombinant zoster vaccine in patients with solid tumors, vaccinated before or during chemotherapy: A randomized trial. Cancer 125, 1301–1312 (2019).

19. Benhamouda N. et al. Cytokine profile of anti-spike CD4+T cells predicts humoral and CD8+T cell responses after anti-SARS-CoV-2 mRNA vaccination. iScience 27, 110441, August 16 (2024).

20. Hirzel C. *et al*. Safety and immunogenicity of adjuvanted recombinant subunit herpes zoster vaccine in lung transplant recipients. *Am J Transplant*. 21(6):2246-2253 (2021).

21. Koldehoff M. *et al*. Cellular Immune Response after Vaccination with an Adjuvanted, Recombinant Zoster Vaccine in Allogeneic Hematopoietic Stem Cell Transplant Recipients *Vaccines*. May 20;10(5):809 (2022).

22. Cheung, Y. K. Dose Finding by the Continual Reassessment Method. New York: Chapman & Hall/CRC Press. (2011).

23. Lee SM, Cheung YK. Calibration of prior variance in the Bayesian continual reassessment method. Stat Med. 30(17):2081-2089. doi:10.1002/sim.4139 (2011)

24. Gould AL. Interim analysis for monitoring clinical trials that do not materially affect the type I error rate. Statistics in Medicine 1992; 11:53-66.

25. Friede T, Kieser M. Sample size recalculation in internal pilot study designs: A review. Biometrical Journal 2006 ;48 :537-555.

26. Birkett MA, Day SJ. Internal pilot studies for estimating sample size. Statistics in Medicine 1994; 13:2455-2463.

27. Gould L. Sample-size re-estimation: recent developments and practical considerations. Statistics in Medicine 2001; 20:2625-2643.

28. Proschan MA. Two-stage sample size re-estimation based on a nuisance parameter: A review. J Biopharmaceutical Statistics 2005;15: 559-574.

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18. LIST OF ADDENDA

18.1. List of investigators

18.2 Serious Adverse Events notification form

18.3 Pregnancy notification form

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